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Paleomagnetism of Triassic and Jurassic red bedded chert of the Inuyama area, central Japan

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Abstract. A paleomagnetic study was conducted on Triassic and Jurassic red bedded chert from the Inuyama area, central Japan. Paleomagnetic specimens were collected from single chert beds at nine sites within mesoscopic (1~5 m) folds, whose ages were determined by radiolarian fossils. Progressive thermal demagnetization experiments were conducted, and three of the four magnetization components provided tectonic implications. Component D (unblocking temperature of 560°-680°C), from six sites, passes the fold test with the mesoscopic folds. Paleolatitude was estimated as 12.3° ± 15.6°N or S, 19.0° ± 10.5°N or S, and 25.7° ± 16.7°N or S for 237 Ma, 223.5 Ma, and 210 Ma, respectively. In the case of northern latitude, a northward migration of 5.6 ± 4.9 cm/yr is estimated between 237 Ma and 210 Ma, with no significant latitudinal migration from 210 Ma to 160 Ma. Component C (unblocking temperature of 350°-530°C) has normal polarity and is thought to have been acquired as a chemical remanent magnetization carried by magnetite, which was acquired as synfolding magnetization with the Sakahogi Synform. The magnetization is considered to have been acquired through heating in the accretionary wedge during the Cretaceous Normal Superchron (120.5-83.5 Ma). Component B (unblocking temperature of 200°-350°C) has reversed polarity and is thought to have been acquired as a thermoviscous remanent magnetization after the bending of the Sakahogi Synform. The most probable period of the magnetization acquisition is between C27r and C24r (58-63 Ma) after the emplacement of Late Cretaceous granitic batholiths.

1. Introduction

Jurassic accretionary complexes are widely distributed among the Japanese Islands, which were formed along the Asian continental margin in Jurassic time. Their stratigraphies have been clarified mainly by using radiolarian and conodont fossils. The ages of these formations generally range from the upper Carboniferous to Jurassic. They were originally formed as the oceanic plate stratigraphic sequence, which is composed, in ascending order, of oceanic basalts, pelagic cherts, hemipelagic claystones, and terrigenous sediments of coarse mudstones and sandstones. It is very important that a major part of the geologic history of such complexes is recorded in cherts that had drifted with oceanic plate movement from a pelagic area to the continental margin. If we could detect the depositional paleolatitude from these cherts by the paleomagnetic method, the movement of the extinct oceanic plate could be reconstructed and the depositional area of the cherts could be considered.

One of the typical Jurassic accretionary complexes along the western margin of the Pacific is distributed in the Inuyama area, Mino belt, central Japan. It is characterized by stacked thrust sheets of chert and clastic rock sequences [*Yao et al.*, 1980; *Kimura and Hori*, 1993]. Since the original stratigraphy has been kept within a thrust sheet of chert sequences, the radiolarian biostratigraphy of the sequence can be established, and it ranges from the Lower Triassic to Lower Jurassic [*Yao et al.*, 1980; *Yao*, 1982; *Hori*, 1990; *Matsuoka et al.*, 1994; *Sugiyama*, 1997].

A rock magnetic study of red cherts was carried out by *Sasajima* [1982], and paleomagnetic studies of bedded chert from the Inuyama area were performed by *Shibuya and Sasajima* [1986]. In the latter study the authors detected a postdepositional remanent magnetization with a positive fold and reversal test and calculated a paleolatitude of 0.7° ± 3.4°N or S. This paleolatitude, in conjunction with that obtained by *Hattori* [1982], has been frequently referred to for the paleogeographic reconstruction of the terrane constituting the Japanese Islands and for estimating the plate motion at that time [e.g., *Maruyama and Seno*, 1986; *Maruyama*, 1997]. The reason is that no Triassic oceanic crust is preserved in the present Pacific, and a Jurassic reconstruction of the Pacific is hampered by the low intensity of magnetic anomalies [*Nakanishi et al.*, 1992a].

Maruyama and Seno [1986] carried out a paleogeographic reconstruction of the Japanese Islands based on the plate reconstruction by *Engebretson et al.* [1985] and geological evidence from Japan itself. For the Mesozoic reconstruction, they also used the estimates of equatorial paleolatitude for the Japanese Islands from Jurassic clastic rocks [*Hattori*, 1982] and Triassic red cherts [*Shibuya and Sasajima*, 1986]. *Maruyama et al.* [1997] produced a schematic reconstruction of the Japanese Islands from 750 Ma to the present. Their figures for the Mesozoic are strongly dependent on the paleogeo-

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graphic reconstruction by *Maruyama and Seno* [1986], and hence dependent on the plate reconstruction by *Engebretson et al.* [1985]. *Hattori*'s [1982] estimates of paleolatitude are based on the measurements of remanent magnetization after progressive alternating field demagnetization (PAFD) at only one step or without any demagnetization and therefore are not reliable. The paleolatitude obtained by *Shibuya and Sasajima* [1986] is based on a high-T unblocking component deduced from thermal demagnetization experiments and is a key datum for the reconstruction of the Japanese Islands in the Mesozoic. A reliable estimate of the paleolatitude, including the assessment of inclination shallowing, will throw a new light on the tectonic history of the Japanese Islands.

Several studies have been done on the red cherts of Permian [*Mankinen et al.*, 1996], Jurassic [*Hagstrum et al.*, 1996; *Hagstrum and Murchey*, 1996], and Mesozoic [*Hagstrum and Sedlock*, 1992; *Hagstrum and Murchey*, 1993] sequences distributed along the eastern margin of the Pacific. In spite of the extensive paleomagnetic work on red chert beds which accreted to the North American continent, there were no attempts to assess the inclination shallowing of the characteristic remanence of these chert beds. Furthermore, no paleomagnetic work was done in Japan after *Shibuya and Sasajima* [1986].

The purpose of this study is to present more reliable estimates of the paleolatitude of chert sequences of the Inuyama area. For this purpose, we used the following procedure: (1) sampling from a single chert bed with determination of the detailed stratigraphic position of the measured samples using radiolarian fossils, (2) fold test on each site, and (3) measurement of the magnetic anisotropy in order to assess inclination error. Four characteristic components of magnetization were obtained, and the tectonic implications will be discussed, including the paleolatitude at the time of deposition.

2. Geological Background and Sampling

The Inuyama area (35.4°E, 137.0°N) is located close to the boundary between Aichi and Gifu Prefectures of central Japan (Figure 1). The Kiso river flows through the study area, along which chert and clastic rocks, i.e., siliceous mudstone, siltstone, and sandstone, occur alternately at a scale of several hundred meters. This alternation of chert and clastic rocks is regarded as a pile of stacked thrust sheets due to offscraping accretionary process in Jurassic time [*Kimura and Hori*, 1993]. Six regional-scale thrust sheets have been recognized in the Inuyama Sequence, and these sheets have been thrust over the younger unit to the south. Besides thrust-produced repetition of lithology, the Sakahogi Synform bent the stacked thrust sheets to produce westward plunging axes [Figure 1; *Kondo and Adachi*, 1975].

Biostratigraphic research, mainly using conodont and radiolaria, revealed that the chert sequence of the Inuyama area ranges in age from Early Triassic to Early Jurassic and that the age of the clastic rock sequence is Middle Jurassic [*Matsuoka et al.*, 1994]. The deformational history of the Inuyama Sequence is divided into two stages, both of which predate Late Cretaceous acidic volcanic rocks [*Kimura and Hori*, 1993]. Stage 1 represents accretion-related deformation characterized by progressive deformation, including in-sequence thrusts, out-of-sequence thrusts, and mesoscopic folds. The accretion-related deformation is considered to have developed through subduction of the Izanagi plate beneath the Asian

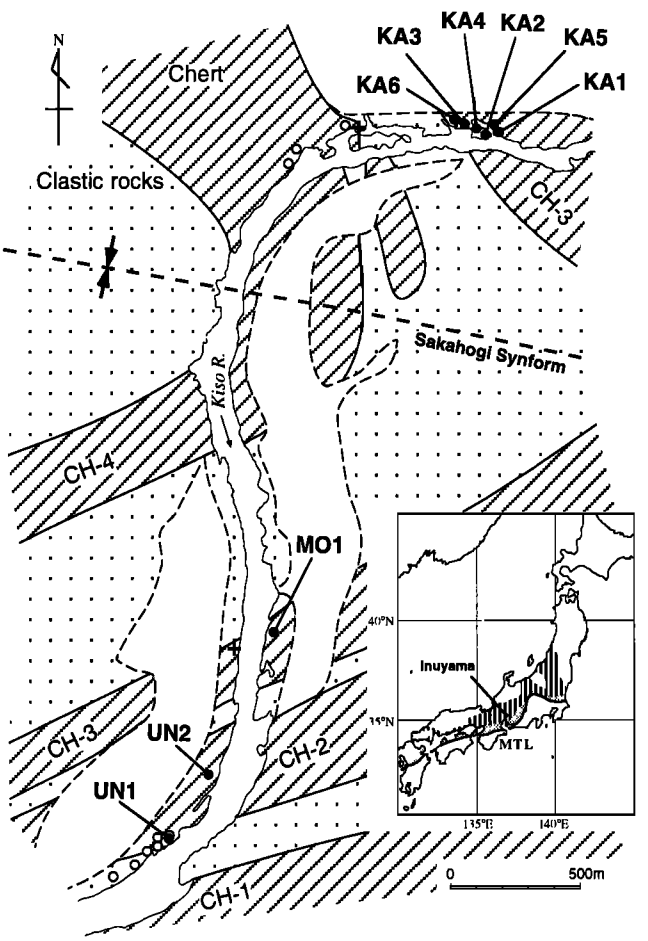


Figure 1. Locality and geologic map of the study area simplified from *Kondo and Adachi* [1975], and *Yao et al.* [1980]. Solid circles are the sampling sites of this study. Open circles are sampling sites by *Shibuya and Sasajima* [1986]. Inset map shows the location of the Inuyama area and the Jurassic accretionary complex of Japan: Mino-Tamba belt (striped), Ryoke metamorphic belt (stippled), and Median Tectonic Line (MTL).

plate during the Jurassic [e.g., *Maruyama*, 1997]. Stage 2 represents intraprism deformation characterized by the development of W-plunging upright folds with wavelengths of 4-12 km including the Sakahogi Synform. They are arranged in a left-stepping mode against the Median Tectonic Line (MTL), which bounds the southern margin of the Jurassic accretionary complexes. They are interpreted to have been formed simultaneously with left-lateral movement along the MTL during the Cretaceous [*Hara et al.*, 1977].

Metamorphic studies of the Jurassic accretionary complex indicate that chaotic units including basaltic rocks experienced the regional metamorphism of prehnite-pumpellyite to pumpellyite-actinolite facies [*Hashimoto and Saito*, 1970]. The Inuyama Sequence, being composed of coherent sedimentary rocks without basaltic rocks, is thought to have been accreted at shallower depths than the chaotic units. *Matsuda and Isozaki* [1991] suggested that the Inuyama Sequence has not been buried below the depth of prehnite-pumpellyite facies metamorphism. After the regional metamorphism during the Jurassic, large granitic batholiths intruded during the Late Cretaceous as a result of the subduction of the ridge between

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the Izanagi and Pacific plates [*Nakajima et al.,* 1990; *Kinoshita,* 1995]. The region heated by the granitic batholiths, known as the Ryoke metamorphic belt, extends along the northern side of the Median Tectonic Line. The metamorphic belt is divided into chlorite, biotite, cordierite, and sillimanite zones. The maximum temperature attained is estimated to be about 700°C. Although the Inuyama area is out of the Ryoke metamorphic belt, the closest granitic batholith is about 40 km from the area.

Chert sequences, the main object of the present study, were named CH-1, CH-2, CH-3, and CH-4 from south to north [*Yao et al.,* 1980]. Within a chert sequence a repetition of the same biozones is recognized owing to layer-parallel thrusts [*Yao et al.,* 1980]. The features of chert are summarized as follows: (1) red or pale green in general; (2) bedded with a bed thickness range of 1.5 to 15 cm; (3) intercalating siliceous claystone layers of 1 to 20 mm in thickness; (4) dipping steeply, in general 60° to 90°; (5) sometimes folded in mesoscopic scale (Figure 2; half wavelengths ranging from 20 to 500 cm [*Kimura and Hori,* 1993]; and (6) mostly including thin quartz veins, generally several hundred micrometers in thickness.

Kimura and Hori [1993] reported the detailed structural features of the mesoscopic folds, including microscopic observations. Chert layers are generally 1.1-1.9 times thicker in the hinges than in the limbs of folds. Flexural flow in interbedded mud layers between competent chert layers produced more thickening in the hinge regions. Hinge areas are commonly fractured and contain axial plane cleavage and bedding-parallel stylolites. Stilettoes developed during loading in an earlier diagenetic stage, whereas other structures are closely related to flexure-slip folding and flattening normal to the axial surfaces. Tests of radiolaria and sponge spicules are arranged parallel to bedding, whereas they are partly rearranged parallel to axial planes in mudstone layers. Tests of radiolaria are undeformed except in the fold hinges, where they are elongated slightly parallel to axial surfaces.

The red bedded chert was sampled at nine sites along the Kiso River; six at Katsuyama (KA1-KA6), two at Unuma (UN1 and UN2), and one at Momotaro (MO1) sections (Figure 1). All sampling sites were chosen so as to cover both limbs of mesoscopic folds of a single bed (Figure 2). At each site the bed was traced without break by deformation. At each site, nine to 10 hand samples were marked by an orientator with a magnetic compass and taken out of the outcrop by a hammer. The orientation of each sample was corrected for the local declination of this area (-7°). The stratigraphic order of the sites was determined according to the biostratigraphic work of *Matsuoka et al.* [1994], except sites UN1 and MO1. Information on the sampling sites, including the thickness of the chert beds and plunge axis of the mesoscopic folds, is listed in Table 1.

3. Laboratory Procedures

3.1. Radiolarian Biostratigraphy

Chert samples were collected for the radiolarian biostratigraphic work from the same bed as those for the paleomagnetic measurements. Cherts were soaked in about 4% hydrofluorlic acid for 24 hours in order to extract a radiolarian test. The resulting residues were sieved on a 63-μm mesh and analyzed by using a scanning electron microscope (SEM).

3.2. Paleomagnetic Measurements

Several cylindrical specimens (25 mm in diameter, 22 mm in length) were cored and cut from each hand sample, and paleomagnetic measurements were conducted on these specimens. Natural remanent magnetization of the specimens was measured with a 2G Enterprises superconducting rock magnetometer (SRM, model 760) at the Geological Survey of Japan. Three hand samples were selected from each site for a pilot study and cylindrical specimens from each hand sample were subjected to progressive alternating field (AF) demagnetization, with 9-15 steps, up to 80 mT using a 2G demagnetizer in line with the SRM. A few samples were demagnetized by an AF demagnetizer with a three-axes tumbling system at Kyoto University (2G model 600), with 19 steps up to 150 mT.

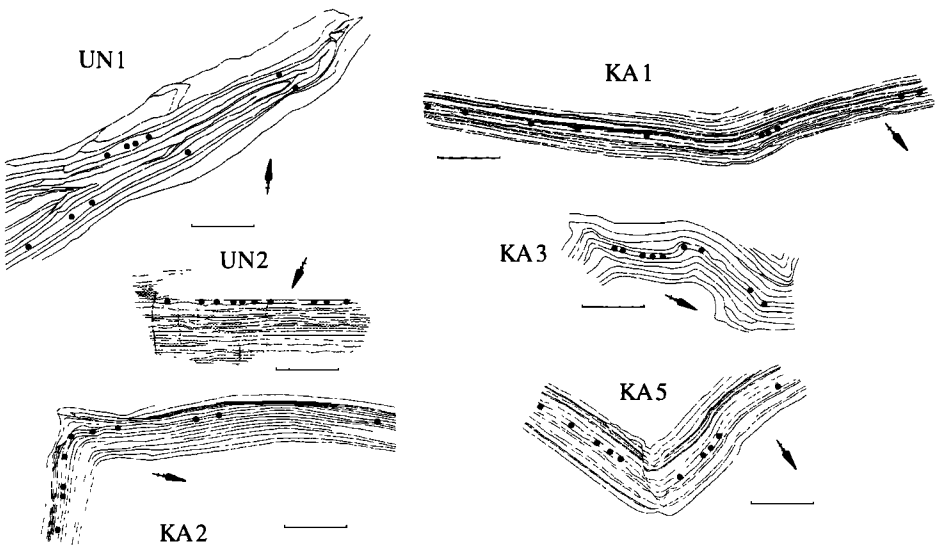


Figure 2. Sketches of outcrops used for this study which passed the fold tests with the mesoscopic folds. Circles are the points where hand samples are taken out. Scale bars are 1 m.

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Table 1. Information of Sampling Sites

Site	Limbs of Sakahogi Synform	Thickness of Single Bed		Fold Axis		Maximum Angle of Rotation
		Total, mm	Clay, mm	Declination	Dip	
UN1	south	80~100	2~20	55.5	4.7	158
UN2	south	30~45	~1	53.3	-18.6	12
KA1	north	15~90	5~60	152	62	37
KA2	north	45~50	1~5	-132.2	70.7	100
KA3	north	40~60	~1	164.3	60.6	61
KA4	north	~50	1~2	176.7	88.2	91
KA5	north	70~85	8~20	-155.4	55.3	74
KA6	north	40~45	1~2	139.9	70	46
MO1	south	~50	~1	62.2	-2.5	71

Fold axis shows a declination and dip of the plunging fold axis of a mesoscopic fold at each site. Maximum angle of rotation is the maximum angle between the bedding planes of the two individual hand samples.

Another cylindrical sister specimen taken from the same hand sample was subjected to progressive thermal demagnetization (PThD) experiments of 17-20 steps from room temperature up to 680°C with a thermal demagnetizer (Magnetic Measurements Co. model MMTD1). The space between the thermal demagnetizer and the mouth of the magnetometer is maintained below ~1000 nT in a magnetic field cancellation cage (Magnetic Measurements Co. model MMLFC) to minimize the acquisition of viscous remanent magnetization (VRM) before measurements. The axes of the specimens were transposed and measured in three different configurations; then the mean value was calculated. According to the results of the pilot PAFD and PThD experiments, specimens from other hand samples were subjected to thermal demagnetization experiments. Low field magnetic susceptibility was measured after each step of the thermal demagnetization with a Bartington Instruments MS2 system to monitor chemical changes of magnetic minerals.

To improve the comparison of PAFD and PThD data, two specimens were selected from each of several hand samples to see the difference in demagnetization character of PAFD after PThD and PThD after PAFD. One is subjected to PAFD up to 80 mT with subsequent PThD up to 700°C, while the second is subjected to PThD up to 530°C with subsequent PAFD up to 80 mT. Magnetic mineralogy was investigated in detail according to the results.

Anisotropy of magnetic susceptibility was measured with a susceptibility anisotropy meter (AGICO Kappabridge KLY-3S) on sister specimens of each sample from almost all the hand samples. Some samples were measured by manual mode in 15 directions and others in rotational mode around three orthogonal axes. Degree of anisotropy (P_J) and shape parameter (T) were calculated by the following formula [*Jelinek*, 1981]:

$$P_J = \exp\{2[(\eta_1 - \eta_m)^2 + (\eta_2 - \eta_m)^2 + (\eta_3 - \eta_m)^2]\}^{1/2} \quad (1a)$$

$$T = [2 \ln (K_{int}/K_{min}) / \ln (K_{max}/K_{min})] - 1, \quad (1b)$$

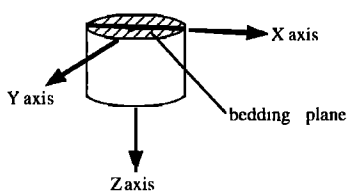
where

$$\eta_1 = \ln K_{max}, \eta_2 = \ln K_{int}, \eta_3 = \ln K_{min},$$

$$\eta_m = (K_{max} K_{int} K_{min})^{1/3}. \quad (1c)$$

Magnetic phase transitions were monitored by using low-temperature magnetic measurements. One hand specimen from each site was crushed in a nonmagnetic tungsten mortar. Powdered samples were subsequently measured with a Quantum Design Magnetic Property Measurement System (MPMS-XL5). Samples were cooled down in zero field to 5 K, at which temperature the 5-T dc field was applied for 60 s and the superconducting magnet was then reset to zero. The isothermal remanent magnetizations were measured at 1° steps during warming up to 300 K.

Stepwise acquisition of isothermal remanent magnetization (IRM) was performed on selected cylindrical specimens at 20 field steps up to 2.5 T with a 2G pulse magnetizer (model 660). With the procedure summarized in Figure 3, the anisotropy of IRM was also evaluated from the IRM acquisition curves to the three orthogonal axes of a specimen. Z is perpendicular to the bedding plane, and X and Y are parallel to



0. Demagnetized (80 mT) state

1. Z axis, 0 -> +2.5 T stepwise IRM acquisition (for checking: Fig. 12)
2. Z axis, 0 > -2.5 T stepwise IRM acquisition (for checking: Fig. 12)
3. Z axis, 0 -> +2.5 T stepwise IRM acquisition (Figs. 12, 13&14)

4. X axis, +2.5 T IRM acquisition
5. X axis, 0 -> 2.5 T stepwise IRM acquisition (for checking)
6. X axis, 0 -> +2.5 T stepwise IRM acquisition (Figs. 13&14)

7. Y axis, +2.5 T IRM acquisition
8. Y axis, 0 -> 2.5 T stepwise IRM acquisition (for checking)
9. Y axis, 0 > +2.5 T stepwise IRM acquisition (Figs. 13&14)

10. Z axis, +2.5 T IRM acquisition
11. Z axis, 0 -> -2.5 T stepwise IRM acquisition (for checking: Fig. 12)
12. Z axis, 0 -> +2 5 T stepwise IRM acquisition (for checking: Fig. 12)

Figure 3. Point form summary of the procedure to obtain anisotropy of isothermal remanent magnetization in Z, X, and Y axes.

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it. The procedure performed here is modified from *Tamai* [1998]. In the approach used by Tamai, for each direction a pulse magnetic field of 2.5 T was given; then the back field from 0 to -2.5 T was applied (dc demagnetization). In the modified approach used here, the dc demagnetization in opposite directions were used to calculate $(IRM_{+2.5} - IRM_H)/2$ (step 2, Figure 3) or $(IRM_H - IRM_{-2.5})/2$ (step 3, Figure 3) to compare with the IRM acquisition curve in the +*Z* direction (step 1; Fig. 3). Here IRM_H is an IRM acquired in the magnetic field of *H* teslas after the application of +2.5 T or -2.5 T. The IRM acquisition curves in the *X* (steps 5 and 6, Figure 3) and *Y* (steps 8 and 9, Figure 3) axes were obtained likewise including the reproducibility check.

Theoretically, these calculated curves are identical to the IRM acquisition curve from the demagnetized state (step 1, Figure 3), provided that the magnetic grains are noninteracting single-domain grains [*Stoner and Wohlfarth,* 1948]. There are some previous works on IRM anisotropy of red beds which contain hematite [*Tauxe et al.,* 1990; *Hodych and Buchan,* 1994; *Tamai,* 1998]. *Hodych and Buchan* [1994] induced the IRM at an angle of 45° to the bedding plane to avoid the problem that IRM tends to be stronger for the first applied direction [*Tauxe et al.,* 1990]. The reproducibility in the *Z* axis, before and after the application of IRM in the *X* and *Y* axes, was also checked (steps 10-12, Figure 3).

Several representative specimens were subjected to the stepwise IRM acquisition up to 2.5 T in the *Z* axis after static alternating field demagnetization (AFD) at 80 mT followed by the stepwise thermal demagnetization (ThD) experiments up to 700°C. After heating at 700°C, stepwise IRM acquisition was conducted and compared with the results before heating to see the change in the coercivity spectrum. After the IRM acquisition in the *Z* axis at 2.5 T, an IRM at 0.3 T was given along the *Y* axis, and an IRM at 80 mT was applied along the *X* axis, successively according to the method by *Lowrie* [1990]. The choice of 80 mT, rather than 100 mT, was because the anisotropy of IRM shows a shoulder at 80 mT, which will be shown later. In order to monitor the thermal decomposition and formation of magnetic phases at the same time, the method modified after *van Velzen and Zijderveld* [1992] and *Torii et al.* [1996] was applied. Before each thermal demagnetization step, the above mentioned orthogonal IRM acquisition was applied, and the change in the composition of magnetic phases was monitored. After each thermal demagnetization step, the magnetization was measured to see the unblocking (plus decomposition). Volume magnetic susceptibility was also monitored during heating, and all the results were used to assess the magnetic mineralogy and the reliability of the magnetization.

4. Results

4.1. Biostratigraphy

In the present study, radiolarian biostratigraphy is used to confirm the stratigraphic horizons from which the paleomagnetic samples were collected. We could identify several index radiolarians from each sample (Table 2) and correlate the horizons as follows:

In KA1, *Triassocampe diordinis* Bragin, *Triassocampe* spp., *Hozmadia* sp., and *Archaeosemantis* sp. were identified. The occurrence of many specimens of *Triassocampe diordinis* and another *Triassocampe* species suggests that radiolarians from KA1 are correlative with the *Triassocampe coronata* assemblage of *Sugiyama* [1992], described from Mount Kinkazan, which is adjacent to the Inuyama area. He correlates the *Triassocampe coronata* assemblage zone to the middle Anisian.

In UN1, *Triassocampe coronata* Bragin was identified. Based on the occurrence of *Triassocampe coronata*, UN1 is also placed within the middle Anisian *Triassocampe coronata* assemblage zone of *Sugiyama* [1992].

In KA5, *Triassocampe deweveri* (Nakaseko and Nishimura), *Pseudostylosphaera japonica* (Nakaseko and Nishimura), and *Oertlispongus* sp. cf. *inaequispinosus* Dumitrica, Kozur and Mostler were identified. Both of the *Triassocampe deweveri* and *Pseudostylosphaera japonica* are included in the *Triassocampe deweveri* assemblage of *Yao* [1982], suggesting a Ladinian age. *Dumitrica* [1982] indicates that the stratigraphic range of the *Oertlispongus inaequispinosus* is from the upper Anisian to lower Ladinian. Consequently, KA5 can be correlated with the lower Ladinian.

In KA2, *Archaeosemantis cristianensis* Dumitrica and *Lysemelas olbia* Sugiyama were identified. According to *Sugiyama* [1997] the horizon yielding both *Archaeosemantis cristianensis* and *Lysemelas olbia* lies within the Norian. Because KA2 is situated stratigraphically between KA5 and KA4 in the KA section, lower Norian age is most plausible for KA2.

In KA4, *Capnuchosphaera* sp. and *Palaeosaturnalis bifidus* (Kozur and Mostler) were identified. *Yao* [1982] reports capnuchosphaerids from the *Triassocampe nova* assemblage of the Inuyama area, suggesting Carnian to middle Norian age. On the other hand, *Kozur and Mostler* [1972] described *Palaeosaturnalis bifidus* under the genus *Spongosaturnalis* from the upper Norian Pötschenkalke of Austria. According to *Yao* [1982], *Palaeosaturnalis bifidus* is included in the *Canoptum triassicum* assemblage of the Inuyama area. Thus KA4 lies on the upper part of the *T. nova* assemblage zone or on the lower part of the *Canoptum triassicum* assemblage zone, correlating with the middle to upper Norian.

In KA3, *Canoptum* sp., *Livarella gifuensis* Yoshida, and *Deflandrecyrtium* sp. were identified. The occurrence of these species suggests that KA3 lies in the *Canoptum triassicum* assemblage zone of *Yao* [1982]. He correlates the *Canoptum triassicum* assemblage zone with the upper Norian to Rhaetian.

In UN2, *Canoptum* sp., *Dumitricaella parva* Sugiyama, *Livarella* sp., *Lysemelas olbia* Sugiyama, and *Palaeosaturnalis* sp. cf. *bifidus* (Kozur and Mostler) were identified. *Canoptum* sp., *Livarella* sp., and *Palaeosaturnalis* cf. *bifidus* are included in the *Canoptum triassicum* assemblage of *Yao* [1982], which indicates upper Norian to Rhaetian age. *Dumitricaella parva* and *Lysemelas olbia* from the Inuyama area were recently described by *Sugiyama* [1997]. Stratigraphic distribution of these species is restricted within the Norian according to *Sugiyama* [1997]. Consequently, UN2 can be correlated with the upper Norian.

In MO1, *Livarella* sp. cf. *gifuensis* Yoshida and *Palaeosaturnalis bifidus* (Kozur and Mostler) were identified. These species belong to the *Canoptum triassicum* assemblage of Yao [1982], suggesting upper Norian to Rhaetian age.

In KA6, *Bipedis fannini* Carter, *Gigi fustis* De Wever, *Praeconocaryomma immodica* Pessagno and Poisson, *Sphaerostylus* sp., and *Parahsuum* spp. were identified.

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Table 2. Results of radiolarian biostratigraphy

Site	Determined Radiolaria	Correlation With Previous Works	Age Attribution
KA6	*Bipedis fannini* *Gigi fusus* *Praeconocaryomma immodica* *Spaerostylus* sp. *Parahsuum* spp.	*Parahsuum simplum* zone [Yao, 1982] *Katroma kurusuensis* subzone [Hori, 1990]	Sinemurian-early Pliensbachian
KA3	*Canoptum* sp. *Livarella gifuensis* *Deflandrecyrtium* sp.	*Canoptum triassicum* zone [Yao, 1982]	upper Norian-Rhaetian
MO1	*Livarella* cf. *gifuensis* *Palaeosaturnalis bifidus*	*Canoptum triassicum* zone [Yao, 1982]	upper Norian-Rhaetian
UN2	*Canoptum* sp. *Dumitricaella parva* *Livarella* sp. *Lysemelas olbia* *Palaeosaturnalis* cf. *Bifidus*	*Canoptum triassicum* zone [Yao, 1982]	upper Norian
KA4	*Capnuchosphaera* sp. *Palaeosaturnalis bifidus*	Upper part of *Triassocampe nova* zone [Yao, 1982] or lower part of *Canoptum triassicum* zone [Yao, 1982]	middle to upper Norian
KA2	*Archaeosemantis cristianensis* *Lysemelas olbia*	TR7 to 8B [Sugiyama, 1997]	lower Norian
KA5	*Triassocampe deweveri* *Pseudostylosphaera japonica* *Oertlispongus* cf *Inaequispinosus*	*Triassocampe deweveri* zone [Yao, 1982]	lower Ladinian
KA1	*Triassocampe diordinis* *Triassocampe* spp. *Hozmadia* sp. *Archaeosemantis* sp.	*Triassocampe coronata* zone [Sugiyama 1992]	Anisian
UN1	*Triassocampe coronata*	*Triassocampe coronata* zone [Sugiyama 1992]	Anisian

Praeconocaryomma immodica was first described by *Pessagno and Poisson* [1981] from the upper Sinemurian or lower Pliensbachian of Turkey below the ammonite-bearing horizon of upper Pliensbachian. After *Hori* [1990] the horizon yielding both *Gigi* and *Bipedis* species is restricted within its subzones I and II, i.e., *Parahsuum* aff. *longiconicum* and *Katroma kurusuensis* subzones which are correlated with Hettangian/Sinemurian and Sinemurian/early Pliensbachian, respectively. Taking the occurrence of *Praeconocaryomma immodica* into account, we correlate KA6 with Hori's subzone II, the Sinemurian to early Pliensbachian.

The stratigraphic horizon of each site determined by radiolarian biostratigraphy is plotted in Figure 4.

4.2. Paleomagnetism

Generally, the PAFD results show three to four components of magnetization, although 10-30% of the natural remanent magnetization (NRM) remained after AF cleaning at the maximum level of 80 mT (Figure 5a). These multiple components have overlapping coercivities showing the trajectory smoothly changing between components. NRM intensity decay curves show a kink at around 35 mT, and above that level the magnetization becomes more difficult to remove by AF. The specimens of sites UN1 and KA1 show a small drop at 5-10 mT AF indicating the presence of low-coercivity components in the sample.

Four components are separated by PThD experiments (Figure 5b). These are characterized as components A, B, C, and D according to *Shibuya and Sasajima* [1986]; their unblocking temperatures were 70°-200°C, 200°-350°C, 350°-530°C, and 560°-680°C, respectively. Components A, B, and C seem to show the same directions as the components observed in the PAFD results. These components are more clearly seen in the PThD results than in PAFD, whereas component D is only partially resolved by PAFD in some specimens. All components are extracted by using principal component analysis [*Kirschvink,* 1980] and average directions are listed in Table 3. Not all samples yielded component D owing to the noisy character of their demagnetization curves.

Susceptibility measured after each heating step generally starts to change above 400°C. It first decreases gradually up to 550°-600°C and then increases again to 680°C. The decrease from 400°C to 550°-600°C may indicate oxidation of magnetite to hematite, and the increase from 550°-600°C up to 680°C may arise from the production of fine-grained magnetite or hematite from the chert matrix. Generally speaking, samples from the same site showed similar behavior. Samples from site KA1 showed significant increase to about 3-9 times the initial susceptibility, while samples from site MO1 showed a significant decrease to 30-40% of the initial susceptibility after heating up to 680°C. The difference in the behavior of magnetic susceptibility during heating may be due to differences between sites in the magnetic mineralogy and the oxygen fugacity of the chert matrix.

4.3. Magnetic Mineralogy

Natural remanent magnetization decay curves by PAFD and PThD give us information about the remanence-carrying magnetic minerals in the red chert used in the study. About

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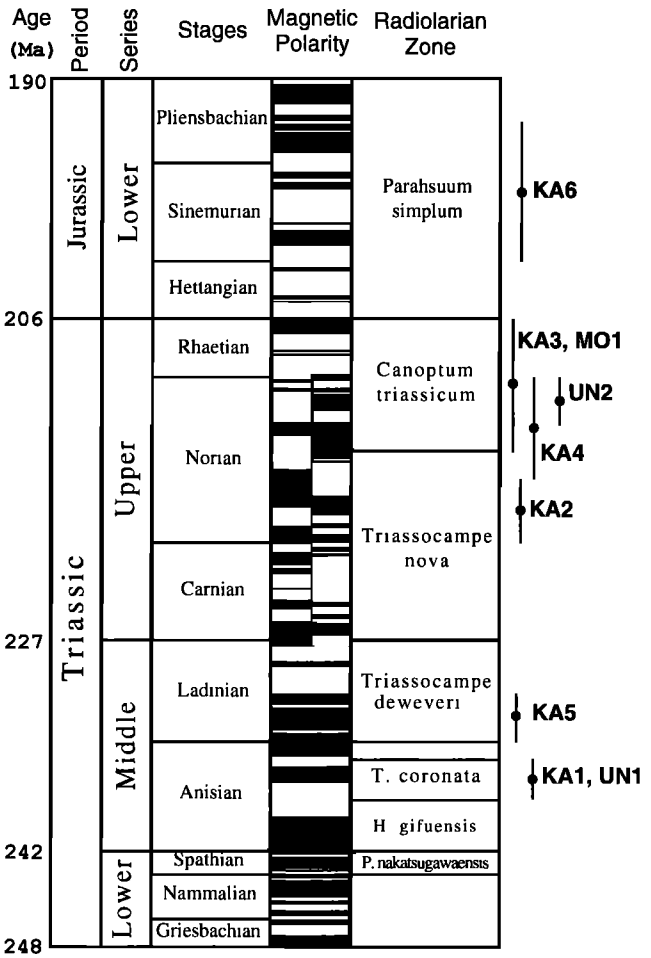
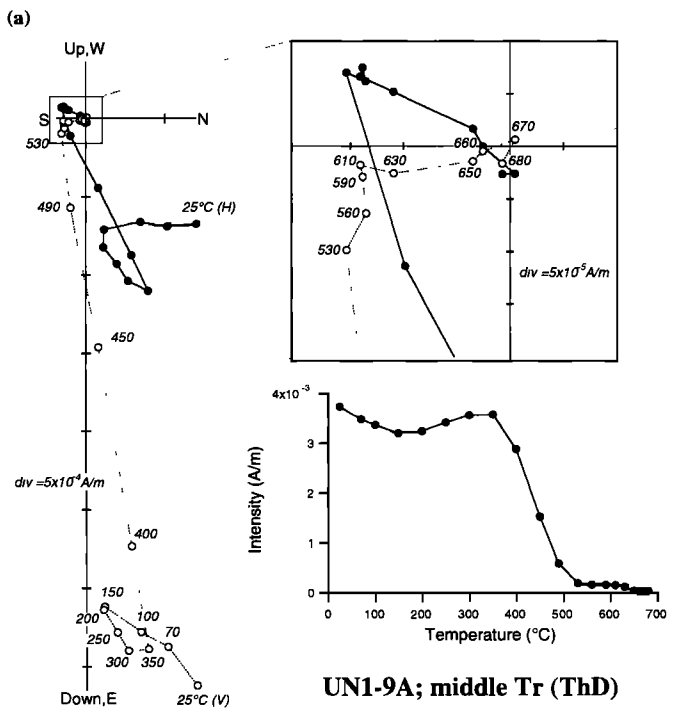


Figure 4. Stratigraphic positions of chert sites based on radiolarian biostratigraphy. Magnetic polarity timescale is from *Gradstein et al.* [1995].

90% of its NRM intensity were lost at around 450°-530°C, and NRM was completely demagnetized at 650°-680°C. NRM sharply decreases to 20-40% of its initial value at around 35 mT of PAFD and then decreases gradually up to the maximum possible field, but it cannot be erased completely. PAFD after PThD and PThD after PAFD was conducted on several hand samples. Figure 6 is one example of the sister specimens from the hand sample KA4-7. PThD up to 530°C erases more than 95% of its NRM (Figure 6d), and subsequent PAFD up to 80 mT could not reduce the intensity of magnetization. This means that the magnetite component was erased by heating to 530°C, but the hematite component remained and survived AFD at 80 mT owing to its higher coercivity. On the other hand, PAFD up to 80 mT eliminate more than 70% of its NRM.



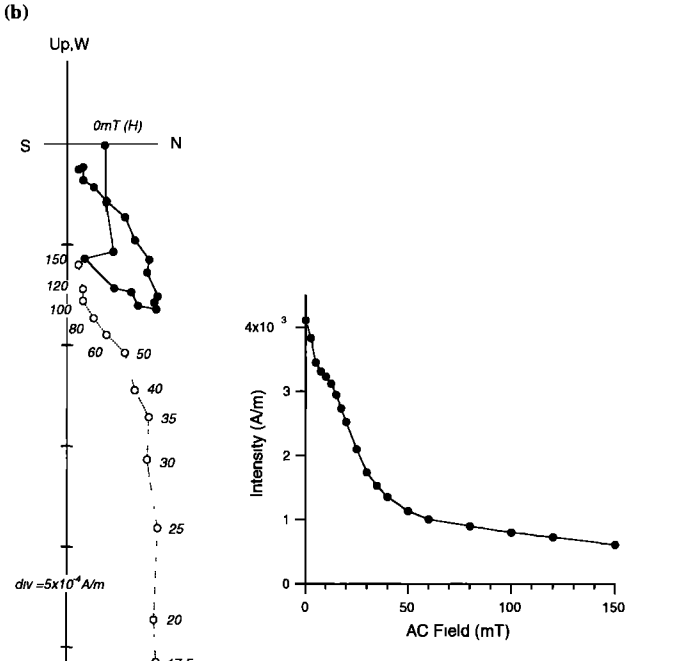


Figure 5. Vector endpoint diagrams of progressive demagnetization experiments on chert samples. Solid circles (open circles) are the magnetization vectors projected onto the horizontal plane (vertical plane). (a) Alternating field demagnetization results for UN1-9B. (b) Thermal demagnetization results for UN1-9A; inset figure is an enlarged version of the vector endpoint diagram to show the region close to origin. Intensities are plotted versus alternating field or temperature.

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Table 3. Paleomagnetic Directions and Progressive Fold Tests Along the Mesoscopic Folds

Component	Site	N	In Situ				Unfolded[a]				Fold Test	Accepted Fold Stages			Maximum Probability			
											Percent Unfold							
			Dec.	Inc.	k	α_{95}	Dec.	Inc.	k	α_{95}	($P > 5\%$)	Pre.	Syn.	Post.	Percent	P	k	α_{95}
A (70°-200°C)	UN1	10	-15 0	62.1	6.0	21.6	-62.3	52.6	3.0	34.1	15~60		Y	Y	40	66	12.2	14.4
	UN2	10	-20.1	30.7	77 4	5 5	35.2	85.1	89 9	5.1	0~100	Y	Y	Y	0	80	89.9	5.1
	KA1	10	-3.7	45.9	13 5	13.6	61.3	47.4	18 8	11.4	5~100	Y	Y	Y	100	88	18.8	11.5
	KA2	10	-3.9	42.0	11.8	14.7	37.7	48.6	7.0	19.6	0~60		Y	Y	30	14	14.2	13.3
	KA3	9	9.1	47.4	29.9	9.6	67.7	28.3	9.5	17.7	0~5			Y	0	7	29.9	9.6
	KA4	10	-34.0	67.0	18.5	11.5	-4.8	-10.4	6.5	20.5	0~30			Y	10	42	18.8	11.4
	KA5	9	5.0	47.0	37.2	8.6	57.7	62.8	11 1	16.2	10~55		Y	Y	30	96	66.5	6.4
	KA6	10	9.0	44.0	44.3	7.3	59.7	13.5	41 3	7.6	0~75		Y	Y	20	72	54.5	6.6
	MO1	10	-17.0	41.0	63.2	6.1	-59.3	74.8	7.0	19.7	0~20			Y	0	28	63.2	6.1
B (200°-350°C)	UN1	10	-135.6	-62.9	8.3	17.8	178.1	-13.1	2 1	45.2	0~7			Y	0	16	8.3	17.8
	UN2	10	-21.4	-1.3	5.6	22.5	-60.7	58.9	5 2	23.4	0~60		Y	Y	0	9	5.2	23.4
	KA1	10	-49.3	-73.2	12.1	14.5	-170.3	-9.4	12.5	14.2	-				0	2	12.1	14.5
	KA2	10	-69.6	-56.7	14.7	13.0	145.1	-19.4	6 4	20.8	0~80		Y	Y	10	97	14.5	13.1
	KA3	9	-73.8	-55.3	28.8	9.8	-179.8	-43.5	13.0	14.8	0~100	Y	Y	Y	20	90	27.1	10.1
	KA4	10	-114.0	-39.0	22.7	10.4	-138.1	28.2	3.9	28.2	0~15			Y	10	52	25.4	9.8
	KA5	9	-20.0	-48.0	19.3	12.0	152.6	-4.8	22.3	11.1	0~100	Y	Y	Y	60	91	25.5	10.4
	KA6	10	-66.0	-62.0	15.0	12.9	177.8	-9.2	11 7	14.7	0~100	Y	Y	Y	0	7	15.0	12.9
	MO1	10	-44.0	-70.0	14.8	13.0	-94.3	-33.1	3.9	28.4	0~5			Y	0	8	14.8	13.0
C (350°-530°C)	UN1	10	68.4	67.0	391.5	2.4	-5.5	38.4	2.2	43.1	-				0	2	391.5	2.4
	UN2	10	34.2	63.9	379.8	2.5	81.9	39.6	136 3	4.2	0~95	Y	Y	Y	0	89	136.3	4.2
	KA1	10	11.0	67.9	34.4	8.4	32.1	35.8	21.8	10.6	0~90	Y	Y	Y	10	30	34.7	8.3
	KA2	10	-8.1	49.5	208.0	3.4	29.5	39 4	8.9	17.1	-				10	3	273.5	2.9
	KA3	9	-5.4	57.1	81.0	5.8	57.7	19.2	14.3	14.1	0~5			Y	0	9	81.0	5.8
	KA4	10	-16.0	54.0	279.2	2.9	-10.9	-26.8	7.0	19.7	0~5			Y	0	58	279.2	2.9
	KA5	9	-18.0	62.0	105.8	5.0	35.3	42.3	6.7	21.5	0~5			Y	0	10	105.8	5.0
	KA6	10	4.0	52.0	62.5	6.2	53.6	7.7	44.1	7.4	-				50	3	73.2	5.7
	MO1	10	44.0	58 0	104.5	4.7	37.2	52.3	15.8	12.5	10~35		Y	Y	20	57	331.3	2.7
D (560°-680°C)	UN1	9	-124.8	-17.8	2.7	39.2	154.7	12.0	3.2	34.3	0~100	Y	Y	Y	100	81	3.2	34.3
	UN2	7	-39.3	-9.7	9.8	20.3	-88.8	46.6	9.9	20.2	0~100	Y	Y	Y	0	91	9.8	20.3
	KA1	10	-35.1	8.2	4.2	26.8	111.9	22.6	4.5	25.9	0~100	Y	Y	Y	80	100	4.5	25.8
	KA2	9	-9.3	16.8	4.8	26.1	82.4	39.4	8.1	19.3	20~100	Y	Y		70	93	9.3	17.8
	KA3	7	4 4	-13.8	2.5	48.5	132.7	16.5	2.9	43.3	0~100	Y	Y	Y	100	56	2.9	43.3
	KA4	9	-57.9	57.0	21.5	11.4	-26.4	-12.3	5.1	23.7	0~15			Y	0	53	21.5	11.4
	KA5	8	-32.0	17.0	8.8	19.8	94.5	29.4	12 5	16.3	13~100	Y	Y		60	66	17.4	13.7
	KA6	3	7.0	42.1	15.3	32.7	-	-	-	-	-	-	-	-	-	-	-	-
	MO1	6	35.2	35.8	18.4	16.0	-22.0	38.8	4.8	24.6	0~70			Y	0	75	18.4	16.0

N is the number of samples measured Dec. is declination, and Inc. is inclination; k is a precision parameter and α_{95} is a 95% confidence limit of the mean directions. Unfolding was conducted along the mesoscopic folds at each site with successive percent unfold according to *McFadden* [1998]. Percent unfold with the probability (P) more then 5% is shown. The accepted stages are shown as Y (for yes), which are defined as Post. (postfolding, 0-10%), Syn. (synfolding, 10-90%), and Pre. (prefolding, 90-100%). The percentages at which the maximum probability was attained are listed with the precision parameters and 95% confidence circles.

[a]Unfolding was conducted along the mesoscopic fold at each site after tilting back the plunged fold axis horizontally.

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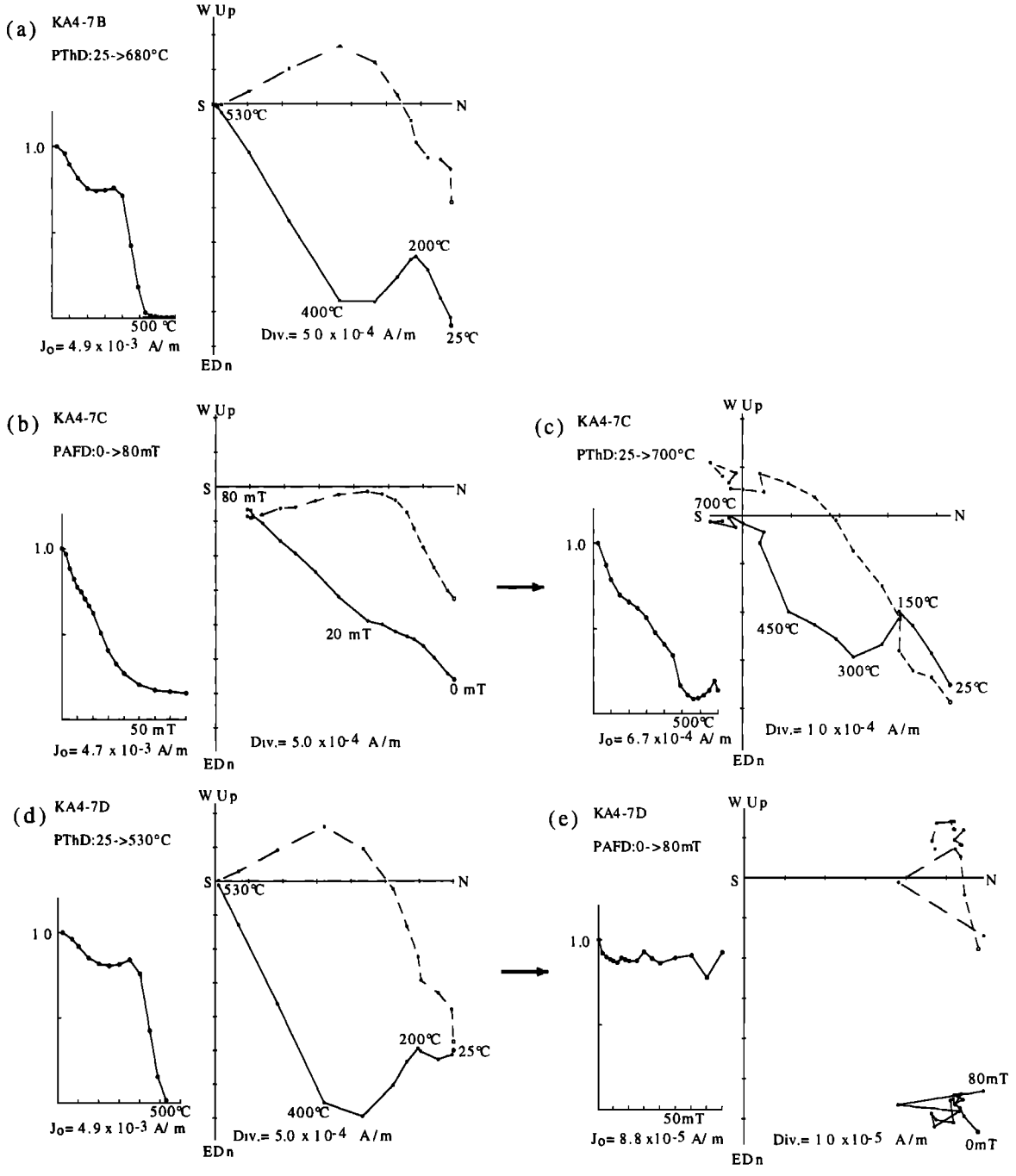


Figure 6. Vector endpoint diagrams of (a) PThD and (c) PThD after (b) PAFD up to 80 mT and (e) PAFD after (d) PThD up to 530°C for the sister specimens from hand sample KA4-7.

The unblocking curve of subsequent PThD (Figure 6c) shows that some amount of magnetite remains with unblocking temperature ranging from room temperature to 530°C. The vector endpoint diagrams show that all four components of magnetization observed in the normal PThD results remain after AFD at 80 mT (Figure 6a), although the intensity of magnetite components was reduced to some extent. They are consistent with the magnetic mineralogy suggested by *Shibuya and Sasajima* [1986].

Magnetic mineralogy is also checked for one sample per site by a low-temperature transition observed during zero-field warming of the remanent magnetization acquired under 5-T dc field at 5 K (Figure 7). Magnetization of the specimens from all sites more or less shows a drop at around 115 K, which is an indication of the Verwey transition characteristic of nearly pure magnetite [*Özdemir et al.*, 1993]. For sites UN2, KA1, KA3, KA4, KA6, and MO1 the transition is clear, whereas it is vague for sites UN1, KA5, and KA2. No indication was obtained for the Morin transition, which is expected to be around 258 K [*Liebermann and Banerjee*, 1971]. This is probably because the grain size of the hematite is too small to show the Morin transition, or impurity has suppressed the transition, which is usually the case for natural hematite.

IRM acquisition curves confirm that both magnetite and hematite occur in the red chert specimens (Figure 8). The IRM of magnetite generally reaches saturation at 300 mT, but that

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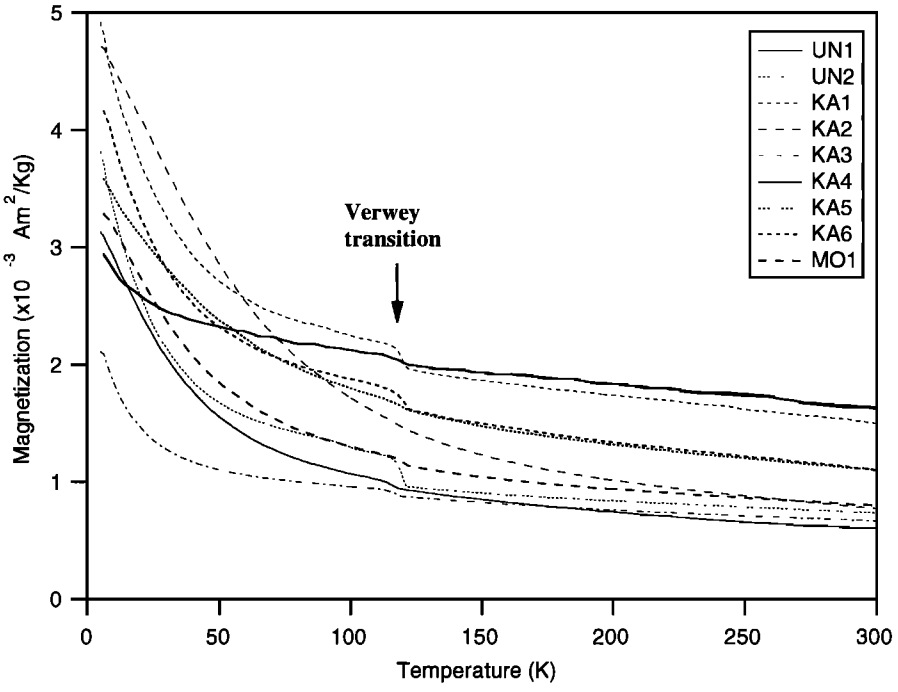


Figure 7. Magnetization versus temperature in a zero field during warming of isothermal remanent magnetization acquired at 5 K at a speed of 2°C/min. Note the Verwey transition observed at around 115 K.

of fine-grained hematite sometimes needs a few teslas for saturation. The observed curves do exhibit two ramps with coercivities corresponding to magnetite and hematite. This can be better viewed as bimodal peaks in the coercivity spectra after differentiating the acquisition curves (Figure 8, shaded areas). The IRM curves seem to show saturation at about 2.5 T.

Part of the evidence of the magnetic mineralogy of the red chert comes from a Mössbauer spectrometry study by *Kubo et al.* [1996] on Triassic bedded chert of various colors from the Inuyama area. They found that the red-colored chert contains hematite with 25-55% of total iron and that purple red samples contain twice as much as brick red samples. They report that the color of each bed is uniform within the bed in general and the color change sometimes occurs at the boundary of the beds. The red color of our bedded cherts indicates the presence of hematite.

4.4. Alteration During Heating

Changes in the magnetic properties including decomposition and formation of the magnetic minerals are investigated by the IRM acquisition experiments before, during, and after heating up to 700°C. The curves of stepwise IRM acquisition experiments before (solid circles, Figure 9a) and after (open circles, Figure 9a) heating show that a significant quantity of magnetic minerals were formed during heating up to 700°C. Coercivity spectra before (solid line) and after (dotted line) heating indicate that both magnetite and hematite were formed. The peak coercivity of magnetite after heating (40-60 mT) is slightly higher than that before heating (10-40 mT). The peak coercivity for hematite is similar before and after heating (about 500 mT), although, the width of the peak is broadened after heating and shows a slight shift to higher coercivity.

In order to see the change in the magnetic mineralogy and the unblocking of each temperature component, orthogonal IRM was imparted at each heating step up to 700°C, and the magnetization before and after heating was measured (Figures 9b, 9c, and 9d). *X* (Figuer 9b), *Y* (Figure 9c), and *Z* (Figure 9d) components represent coercivity fractions of 0-80 mT, 80-300 mT, and 300-2500 mT, respectively. The magnetization after heating (solid circles) at each step is affected by the unblocking of the magnetization at room temperature, decomposed magnetization, and newly formed magnetization that is not unblocked at that temperature. The magnetization before heating (open circles) at each step includes the magnetization at room temperature, decomposed magnetization, and newly formed magnetization.

The results show that higher coercivity components are carried by hematite with a sharp unblocking temperature of 680°C (solid circles, Figure 9d). The magnetization before each heating step (open circles, Figure 9d) shows only a slight change up to 590°C, which suggests that there was no decomposition, nor formation, in this coercivity range. Above 590°C the magnetization before heating step increases, suggesting the formation of hematite. However, the unblocking curve (solid circle, Figure 9d) shows that no hematite survives unblocking at that temperature, which means that there is no significant effect on our PThD experiments. Lower coercivity components (solid circles, Figures 9b and 9c) show unblocking at 580°C and are considered to be mainly carried by magnetite. However, there is a slight indication of hematite in components of both 0-80 mT and 80-300 mT. The component before ThD between 0 and 80 mT shows a slight decrease up to 620°C and hereafter increases (open circles; Figure 9b). The decrease may indicate slight decomposition of magnetite up to 620°C and the formation of magnetite above 620°C. Volume magnetic susceptibility during heating (Figure 9e) shows a decrease above 400°C, followed by an increase above 590°C. This is a combination of ferrimagnetic and paramagnetic minerals, whose

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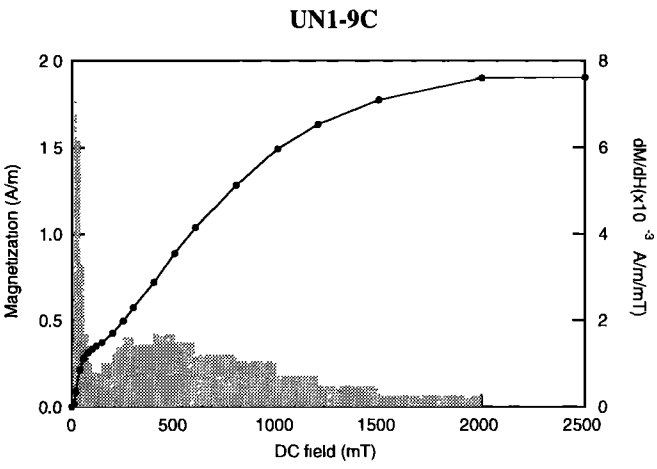


Figure 8. Solid circles represent the IRM acquisition curve from the demagnetized state to +2.5 T. Shaded columns are the derivative of the magnetization, which indicate the unblocking spectrum of the samples.

mechanism is not easy to understand and not direct information of the change in the remanence-carrying minerals. The newly formed magnetite and hematite do not seriously affect the PThD experiments because they have unblocking temperatures at or below the temperature at which they are formed. However, the viscous component of the assemblage may have affected the quality of the data resulting in the noisy results of component D from the sites with a lower hematite abundance.

5. Discussion

5.1. Fold Tests Along Mesoscopic Folds

The components of magnetization calculated are subjected to the progressive fold tests along the mesoscopic folds according to the method proposed by *McFadden* [1998]. This method was extended from the test by *McFadden and Jones* [1981], which imposed a requirement that the bedding planes be consistent within the group. Two clusters were made on each limb of the fold for the analysis. The results are listed in Table 3. Percent unfolding whose probability (P) exceeds 5% is shown as accepted intervals. Prefolding (90-100% unfolding), synfolding (10-90% unfolding) and postfolding (0-10%) magnetizations are accepted if the probability exceeds 5% in the corresponding interval. Percent unfolding at which the maximum probability is attained is also shown with the probability, precision parameters, and α_{95} confidence angles.

Although the number of samples at each site is not enough to give confident results, general behavior can be extracted. The large range of acceptable unfolding percentages, especially for the components B and D, may be related to the large α_{95} associated with the scattered data. For components A, B, and C, the prefolding magnetizations are not accepted in general. Some of the sites accept all the stages from prefolding through synfolding to postfolding magnetization with the probability above 5%. This can be considered to occur because the number of samples and stability of the magnetization is not enough.

For component D, the situation is complicated. For sites KA4 and MO1 the idea that the magnetization was acquired prefolding is rejected and a post-folding magnetization is preferred. For sites KA2 and KA5, the idea that the magnetization was acquired post-folding was rejected. The other four sites (UN1, UN2, KA1, and KA4) exceed the 5% probability criteria for the prefolding, synfolding, and postfolding magnetizations. However, a prefolding magnetization is the preferred option, which can be seen in the percent unfolding of the maximum probability (100-60%). Thus the postfolding magnetizations for sites KA4 and MO1 can be interpreted as incomplete removal of component C. The evidence that the mean angles between the component C magnetizations and component D magnetizations for each specimen are 24.8° and

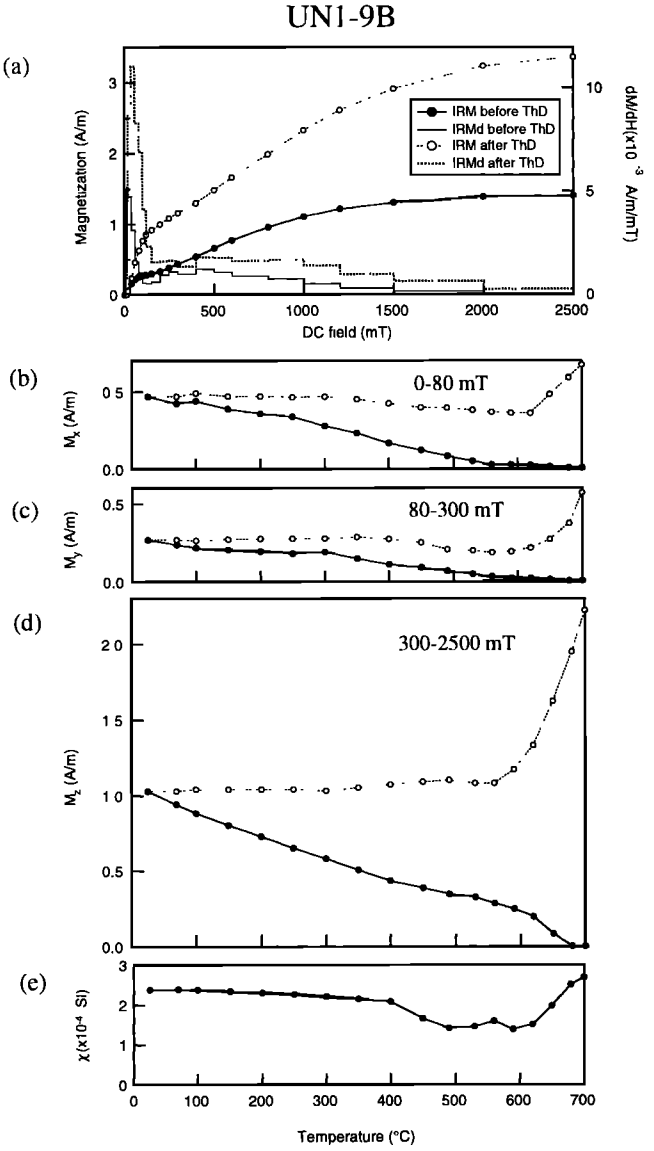


Figure 9. Change in the magnetic properties during heating of the specimen UN1-9B. (a) IRM acquisition curves before (solid circles) and after (open circles) heating up to 700°C in the *Z* axis; Coercivity spectra are also shown as solid (before heating) and dotted (after heating) lines, respectively. At each step before heating, IRM is as follows: (b) 80 mT in the *X* axis, (c) 0.3 T in the *Y* axis, and (d) 2.5 T in the *Z* axis. The magnetizations before (open circles) and after (solid circles) each heating step were measured and plotted. (e) Volume magnetic susceptibility was also monitored at each step during heating.

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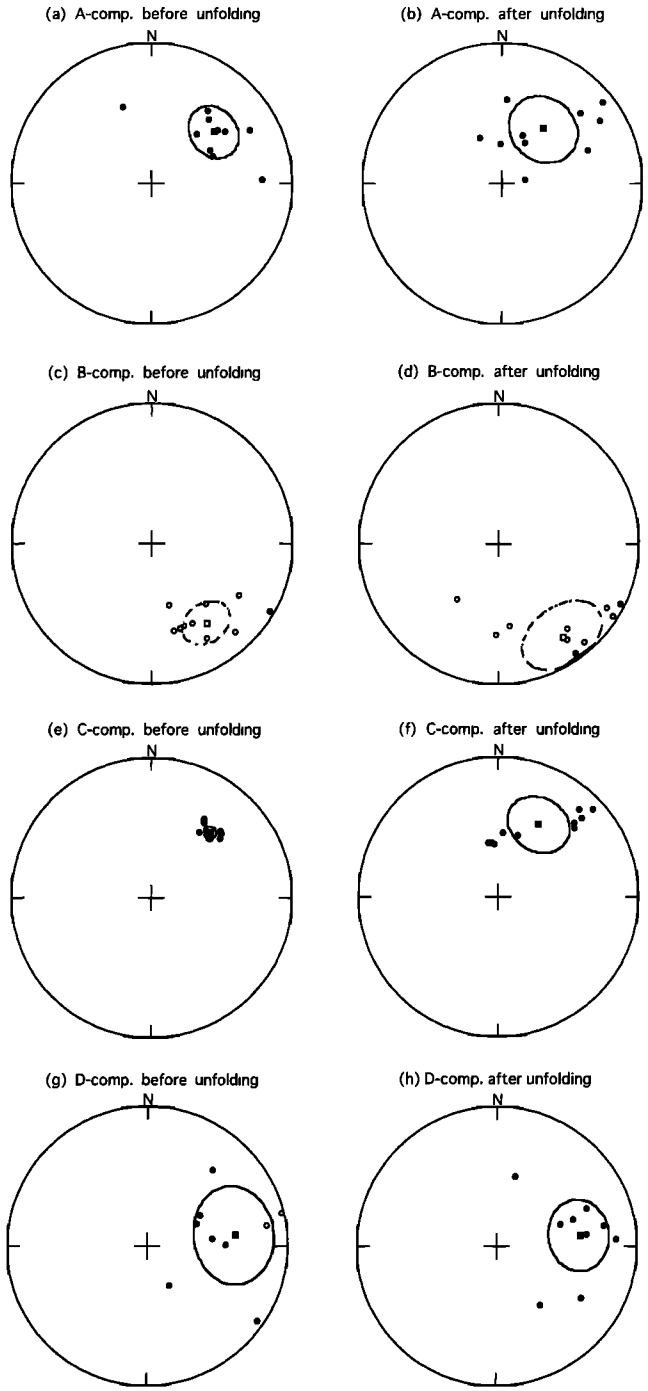


Figure 10. Directions of four components of magnetization for the specimens at site KA2 before (left column) and after (right column) the full unfolding along the mesoscopic fold. Circles represent the directions of the specimens. Mean directions and 95% confidence limits are presented as rectangles and circles, respectively. Solid (open) symbols and solid (broken) curves represent the directions in the Southern (Northern) Hemisphere.

27.8° for KA4 and KA6 may support this idea. Other sites show mean angles of 117.2°, 92.6°, 70.6°, 41.9°, 79.9°, and 49.3° for UN1, UN2, KA1, KA2, KA3, and KA5, respectively. Other evidence that the maximum demagnetization levels used to calculate component D magnetization for KA4 are mostly 620°C, which are relatively low unblocking temperatures, also supports that component D might have been contaminated by component C. Although UN2 shows maximum probability at 0% unfolding, the difference in the precision parameters is only 0.1 between 9.8 (at 0%) and 9.9 (at 100%). We assume that component D of UN2 was acquired before the bending of the mesoscopic folds. In the later discussions, the component D magnetizations, except for sites KA4, KA6, and MO1, are assumed to have been acquired prefolding. The directions of each component for site KA2 are plotted on the Schmidt plot before and after the unfolding of the mesoscopic fold (Figure 10).

Component A has positive inclinations and is close to the present-day axial dipole field directions at the studied area (inclination of +55°). This indicates that component A was acquired after the major tectonic movement of this area ceased, probably as a VRM during the Brunhes Chron. We will discuss the origin of components B, C, and D in the following sections.

5.2. Fold Tests Along the Sakahogi Synform

Shibuya and Sasajima [1986] demonstrated that the two directions of component C from the northern and southern limbs pass the fold test by unfolding the general trend of the Sakahogi Synform through 90° about the vertical axis. *Kimura and Hori* [1993] and *Kimura* [1999] suggested using a fold axis with a trend of N82°W, plunging 78°, and a rotation angle of 110°, to unfold the chert sequence CH-3 of the northern limb to sequence CH-2 of the southern limb (see Figure 1). *Kimura* [1999] has shown that the above mentioned parameters can reconcile the slip vectors of left-lateral faults and the poles of the fault planes in the northern and southern limbs. Thus we have conducted the progressive unfolding

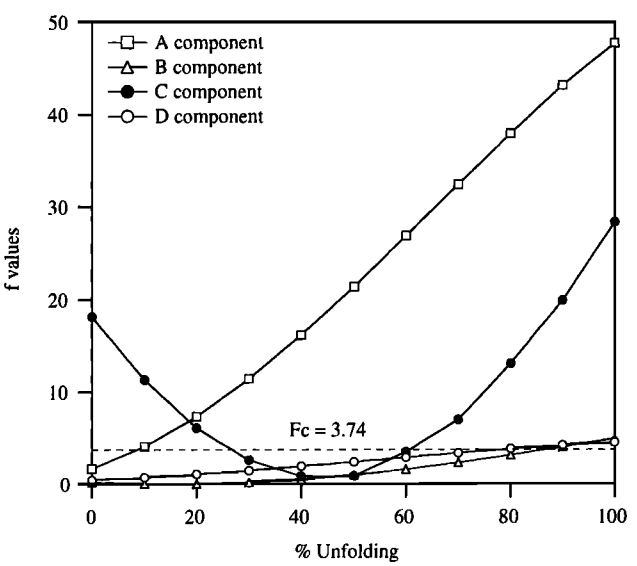


Figure 11. Results of progressive unfolding along the Sakahogi Synform on the insitu directions of the four components listed in Table 3 according to *McFadden and Jones* [1981]. Calculation was made with the mean directions before unfolding of the mesoscopic folds. Mean directions of northern and southern limbs were used for calculation, and f values are plotted versus percent unfolding. Critical value (95% rejection limit) for the data set ($F_C = 3.74$) is shown as a horizontal dashed line.

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along the same fold axis as that of *Kimura* [1999] by fixing the southern limb.

We follow the method of progressive unfolding by *McFadden and Jones* [1981], because only a single bedding plane is used for each limb of the synform and because their method is easier to calculate and understand. All four components were progressively unfolded and f values were calculated and plotted versus percent unfolding (Figure 11). The paleomagnetic parameters for each stage of the four components are listed in Table 4. The unfolding is justified at the 95% confidence limit when the f values are below the critical f value (3.74 in this case). The results show that the hypothesis of a common mean of component C for the two limbs is accepted at the 95% confidence limit between 25% and 60% unfolding and is rejected for other unfolding values. Component A clearly shows the minimum value at 0% unfolding which rapidly increases with increasing percentage of unfolding and goes beyond the critical value. This means that component A magnetization was acquired after the bending of Sakahogi Synform was completed. Values of f for components B and D show only a slight increase with percent unfolding away from zero, which may be a result of the large scatter of the data. These two f values cross the critical value (3.74) only above 80%. This means that 0% unfolding is preferred, but any value between 0 and 80% is possible at the 95% confidence limit. It is strange that for component D the unfolding along the mesoscopic folds suggests prefolding or synfolding magnetization, whereas that along the Sakahogi Synform indicates synfolding or postfolding magnetization. The combination of the two fold systems might have made it possible for component D to appear as synfolding or postfolding magnetization along the Sakahogi Synform. Unfortunately, the exact geometrical relationship between the mesoscopic folds and Sakahogi Synform is not clear. The mean directions of all four components for all the sites are plotted with the 95% confidence circles in Figure 12.

5.3. Anisotropy of Magnetic Susceptibility

The degree of anisotropy (P_J) is plotted versus magnetic susceptibility for all the samples (Figure 13a). Magnetic susceptibility generally ranges from 0.5~7 x 10^{-4} SI and samples from sites UN1, MO1, KA2, and KA5 show higher values (2~7 x 10^{-4} SI) than those from other sites. The shape parameter (T) is plotted versus P_J in Figure 13b. The shape of the anisotropy ellipsoid is prolate when $-1 < T < 0$ and oblate when $0 < T < 1$. Samples from the southern limb of the Sakahogi Synform (sites UN1, UN2, and MO1) show a higher degree of anisotropy (1.03~1.19), mostly oblate, whereas samples from the northern limb show essentially lower anisotropy (1.01~1.07) which is both prolate and oblate. This suggests that the southern limb was more deformed than the northern limb, provided that the magnetic minerals responsible for the anisotropy are the same in both limbs or have the same relationship with strain direction. This may almost be the case, in spite of the slight difference in the coercivity spectra of IRM (see the next section). The oblate shape of the anisotropy for the southern limb suggests that the southern part was flattened in the direction of minimum axes rather than extended in the direction of maximum axes.

The angles between K_{min} axes and the directions of magnetization are plotted versus P_J for component A (Figure 13c), component B (Figure 13d), component C (Figure 13e), and component D (Figure 13f). Component C shows the angles close to 90° for the samples from sites UN1 and MO1 with higher anisotropy degree. This may suggest that the component C directions from these sites are affected by the fabric of anisotropy of magnetic susceptibility (AMS), which may be caused by the flattening of the beds during the deformation by accretion or by the bending of Sakahogi Synform. Other components do not show significant features indicating the correlation between AMS and NRM directions. As for maximum axes of AMS, no correlation was found between P_J and angles to each component of magnetization.

Progressive fold tests along the mesoscopic folds are conducted on the K_{max} and K_{min} axes of AMS for each sites. Because the AMS data are not vector and not Fisher distributed, the bootstrap method by *Tauxe and Watson* [1994] was used for the progressive fold tests. Table 5 shows the results of progressive fold tests. K_{max} and K_{min} axes are yielded to the fold tests separately. K_{max} shows significant results only for sites UN1, KA5, and MO1. For site KA5, postfolding development of the K_{max} axis is preferred, whereas for sites UN1 and MO1, prefolding results are preferred. As for the K_{min} axis, sites UN1, UN2, KA4, KA6, and MO1 give meaningful results. Only sites UN1 and MO1 give significant results for both the K_{max} and K_{min} axes. They

Table 4. Paleomagnetic Mean Directions Through the Unfolding of the Sakahogi Synform

Component	N	Percentage Unfolding	Declination	Inclination	k	α_{95}
A (70°-200°C)	9	0	-6.6	48.2	32.3	9.2
		100	89.4	78.0	4.7	26.5
B (200°-350°C)	9	0	-62.1	-59.0	6.9	21.1
		100	31.0	-52.1	4.1	29.1
C (350°-530°C)	9	0	9.0	61.8	27.8	9.9
		50	51.6	66.7	88.4	5.5
		100	105.1	72.7	19.6	11.9
D (560°-680°C)	9	0	-24.4	19.4	3.0	36.0
		100	81.9	47.9	1.8	55.4

Unfolding was conducted along the Sakahogi Synform with the southern limb fixed. N is the number of sites; k and α_{95} are precision parameter and 95% confidence angle, respectively.

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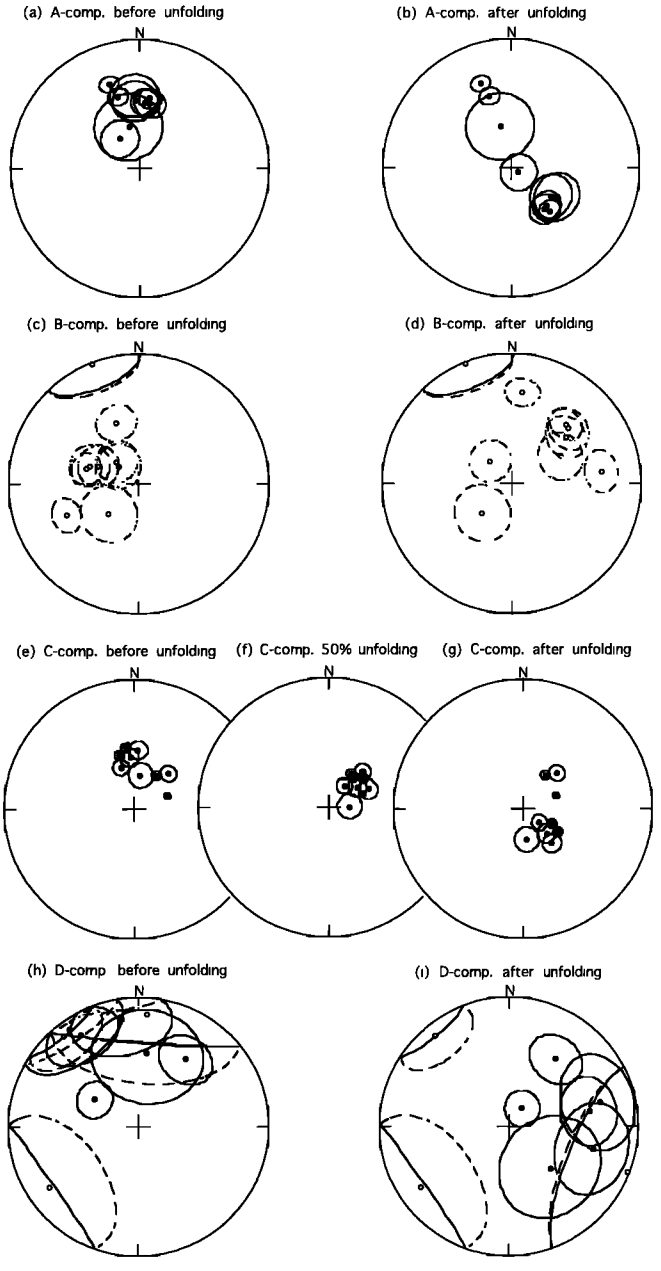


Figure 12. Directions of four components of magnetization before (left column) and after (right column) the full unfolding along the Sakahogi Synform. Solid (open) circles and large solid (dashed) curves represent the directions of the specimens and 95% confidence limits in the Southern (Northern) Hemisphere, respectively. For component C the results of 50% unfolding are also shown.

show a preference for synfolding development (10~50% unfolding) of the K_{min} axis. The results of progressive unfolding for sites UN1 and MO1 are shown in Figure 14. For the plunge of the mesoscopic fold axes, the directions of geographic coordinates were untilted to be horizontal prior to the progressive fold tests; thus the horizontal directions obtained after 100% tilt adjustment mean that the directions are parallel to the bedding plane. K_{max} axes for both sites cluster along the horizontal directions after 100% unfolding, which means the K_{max} directions are aligning within the beds. The reason why only sites UN1 and MO1 show significant results may be that these two sites show a strong degree of anisotropy and that the magnetic fabric was well enough developed to give significant results. These results suggest that the magnetic fabric representing K_{max} axes are developed before the folding of the mesoscopic fold, whereas K_{min} axes are developed during the formation (maybe a later stage) of the mesoscopic folds.

Figure 15 shows the in situ directions of the K_{max} and K_{min} axes of the samples from the sites in the north and south limbs of Sakahogi Synform. For the north limb, no clustering can be observed for either the K_{max} or the K_{min} axes. On the other hand, the K_{min} axis for the south limb shows significant cluster in the horizontal NNW direction. This direction is almost perpendicular to the directions of fold axes for the south limb sites (Table 1). Combined with the analysis along the mesoscopic folds, it can be inferred that the magnetic fabric of the K_{min} axes for sites UN1 and MO1 is developed by shortening, in the direction perpendicular to the mesoscopic fold axes and subhorizontal directions, during the later stage of the formation of the mesoscopic folds.

5.4. Anisotropy of Isothermal Remanent Magnetization

The first IRM acquisition curves from the demagnetized state go slightly higher than those from different initial conditions. However, the IRM acquisition curves from +2.5 T to -2.5 T and from -2.5 T to +2.5 T are nearly the same (Figure 16). This is also true for the IRM imparted along the X and Y axes. Anisotropy of isothermal remanent magnetization (AIR) was estimated by stepwise acquisition of IRM in each axis from the state with the IRM at -2.5 T (Figure 17). In order to discriminate the contribution of low and high coercivity components, IRM_x and IRM_y are plotted versus IRM_z for each step of the IRM acquisition (Figure 18). If there is no anisotropy, they should be perfectly aligned along the line with slope of unity. All plots show two straight line segments separated by an elbow at around 80 mT. The magnetic mineral corresponding to the part with coercivities lower than 80 mT is dominantly magnetite, and that with coercivities higher than 80 mT is mainly hematite. Although there is some fraction of hematite in the 0- to 80-mT component and some magnetite in the 80- to 300-mT component, which can be observed in Figure 9. All calculated slopes are listed in Table step of the IRM acquisition (Figure 18). If there is no anisotropy, they should be perfectly aligned along the line with slope of unity. All plots show two straight line segments separated by an elbow at around 80 mT. The magnetic mineral corresponding to the part with coercivities lower than 80 mT is dominantly magnetite, and that with coercivities higher than 80 mT is mainly hematite. Although there is some fraction of hematite in the 0- to 80-mT component and some magnetite in the 80- to 300-mT component, which can be observed in Figure 9. All calculated slopes are listed in Table 6, together with the results of AMS.

If the primary fabric of the hematite is preserved in the red chert, the IRM acquired along the Z axis should be the smallest and the shape of the ellipsoid should be oblate. The shape of the ellipsoid for the hematite component of KA2-8B is oblate and KA3-8B is triaxial, and that of KA1-7B is nearly isotropic. However, the slope for the hematite component along the X axis is largest for UN1-9C and along the Y axis is largest for KA5-1A with the prolate shape ellipsoid.

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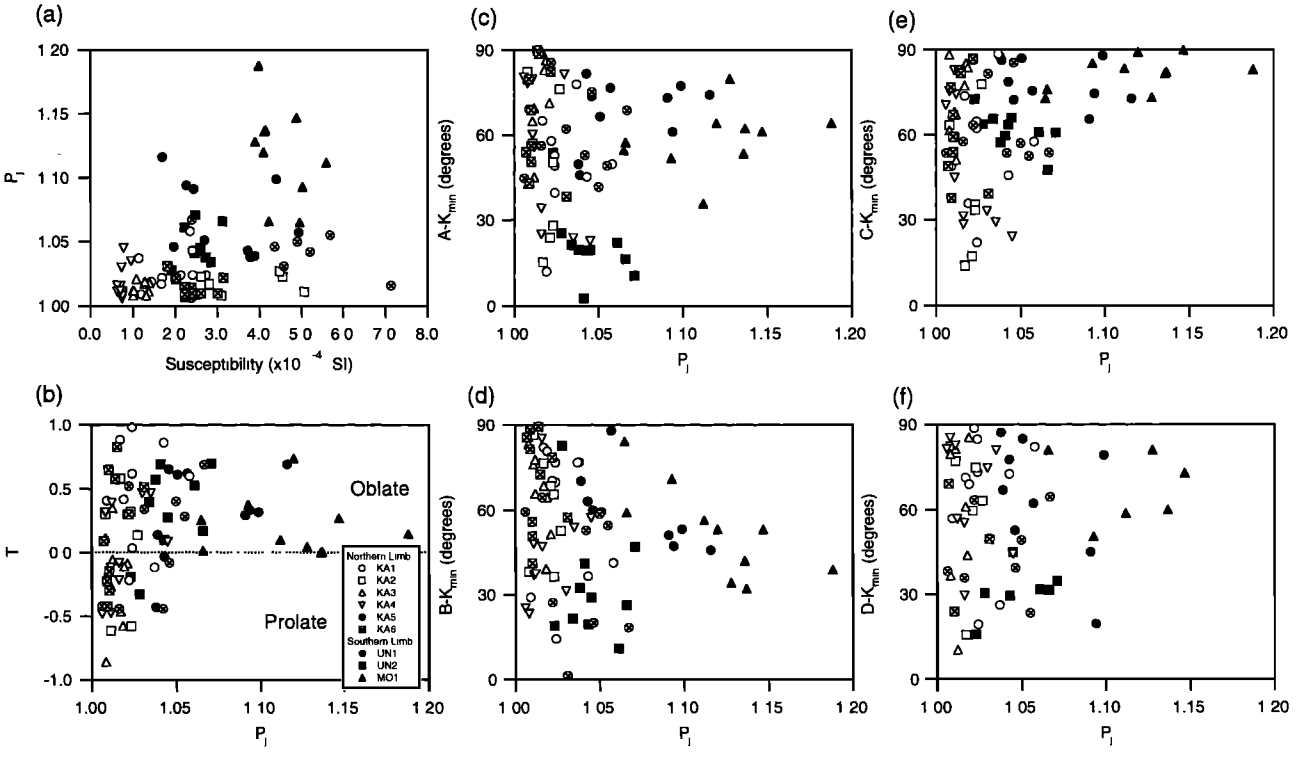


Figure 13. Results of anisotropy of magnetic susceptibility. (a) Degree of anisotropy (P_J) plotted versus magnetic susceptibility. (b) Shape parameter (T) plotted versus degree of anisotropy. Note the larger P_J for the samples from the sites of the southern limb of the Sakahogi Synform (solid symbols), which may indicate stronger deformation. Angles of magnetization directions of components (c) A, (d) B, (e) C, and (f) D to the K_{min} axis directions are also plotted versus P_J.

There are two possible causes for the anisotropy of IRM. One is by compaction during burial of the chert bed at the ocean floor. Another is due to tectonic deformation of the chert sequences during accretion and uplift. The prolate ellipsoid of IRM anisotropy can be considered as the overlapping effect of tectonic deformation on the primary fabric. Difference in shape and degree of anisotropy between the magnetite and hematite component means that the tectonic defor-

Table 5. Progressive Fold Tests Along the Mesoscopic Folds on the Pricipal Directions of AMS

Axis	Site	N	Fold Test (95% fraction)	Accepted Fold Stages		
			Percent Unfold	Prefolding	Synfolding	Postfolding
K_{max}	UN1	10	96~114	Y		
	UN2	10	-37~111	Y	Y	Y
	KA1	10	-60~160	Y	Y	Y
	KA2	7	-60~160	Y	Y	Y
	KA3	9	-8~137	Y	Y	Y
	KA4	10	-11~158	Y	Y	Y
	KA5	9	-47~9			Y
	KA6	10	-60~160	Y	Y	Y
	MO1	10	141~160	Y		
K_{min}	UN1	10	11~33		Y	
	UN2	10	85~160	Y	Y	
	KA1	10	-60~160	Y	Y	Y
	KA2	7	-60~160	Y	Y	Y
	KA3	9	-60~160	Y	Y	Y
	KA4	10	104~144	Y		
	KA5	9	-60~102	Y	Y	Y
	KA6	10	-60~24		Y	Y
	MO1	10	21~51		Y	

Bootstrap fold tests are conducted according to *Tauxe and Watson* [1994].

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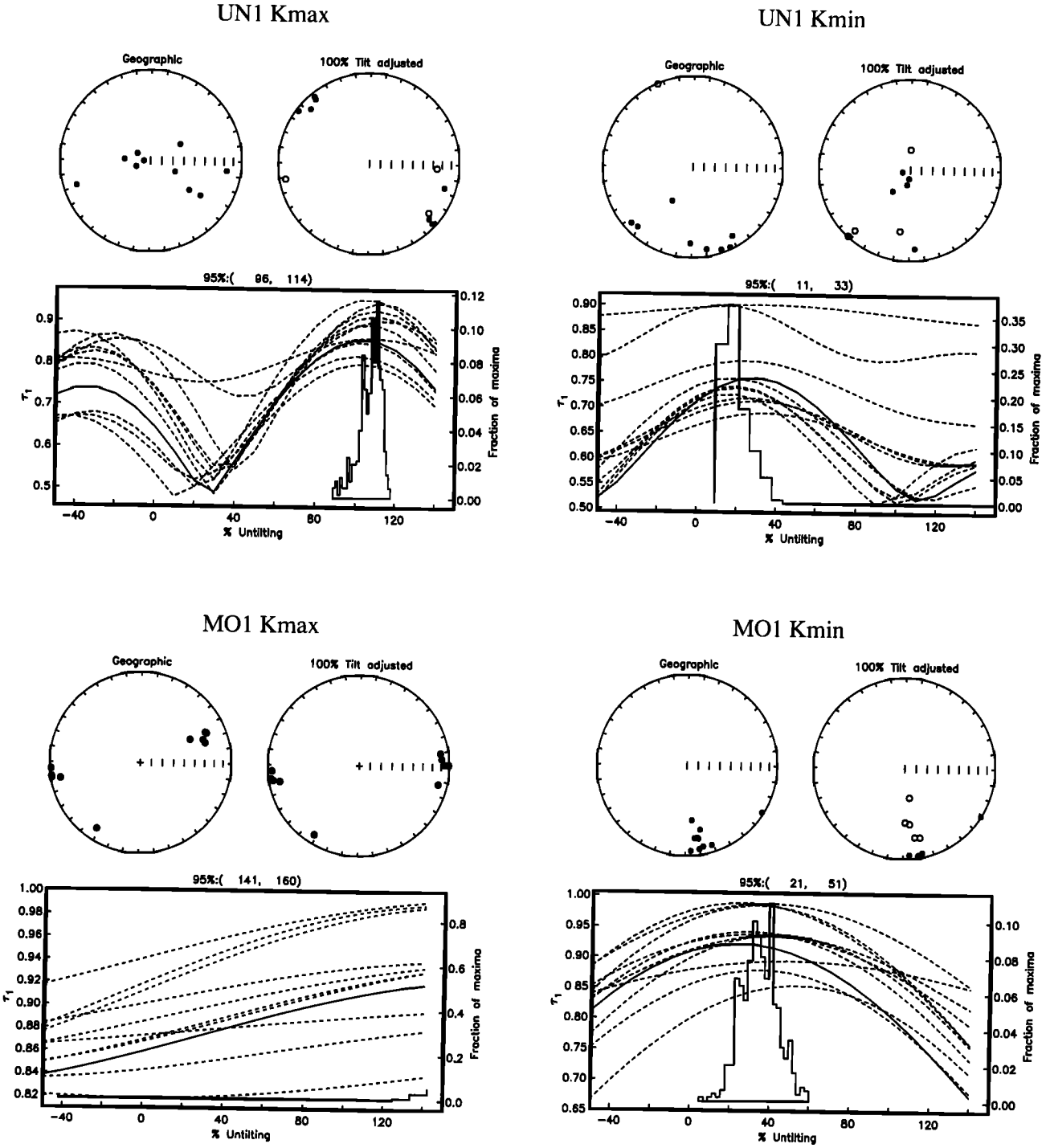


Figure 14. Results of bootstrap fold test [Tauxe and Watson, 1994] on K_{max} and K_{min} directions of sites UN1 and MO1. Prefolding and synfolding fabric is suggested for K_{max} and K_{min} directions for both sites.

mation acted differently on the two minerals. In general, the degree of anisotropy is higher for magnetite than hematite. The individual grain anisotropy of hematite is generally high [$P_J > 100$ [*Tarling and Hrouda,* 1993, Table 2.1]) and can be considered to be nearly infinite, as *Hodych and Buchan* [1994] also assumed. By applying the relationship between ARM anisotropy and inclination shallowing by *Jackson et al.* [1991] to IRM anisotropy, the magnetic inclination at the time of deposition (I_F) and inclination of the remanence (I_{DRM}) can be expressed as

$$\tan I_F / \tan I_{DRM} = IRM_{max}/IRM_{min}, \quad (2)$$

where IRM_{max} and IRM_{min} are maximum and minimum values of IRM, respectively. The use of the equation was suggested by *Hodych and Buchan* [1994], because both the multidomain and single-domain hematite is magnetized in the basal plane and can be a substitution of anhysteretic remanent magnetization of single-domain magnetite.

Although our sample seems to have suffered from tectonic deformation somewhat, the maximum possible shallowing

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factor relative to the bedding plane can be estimated from the relationship by *Jackson et al.* [1991]. In order to correct the inclination, we take the larger of the values I_x/I_z and I_y/I_z as I_{hmax}/I_z. This value is used to correct the inclination by assuming the maximum possible inclination shallowing by equation (2). Results of inclination correction are listed in the right-most column of Table 6. The results show that the correction changes the inclination value by only 2° at most. This means that the inclination error due to compaction and/or tectonic deformation is small. It should be noted that there is a limitation in the above discussion because the AIR was measured only in the three primary directions and the correction is based on the assumption that there is only a flattening factor.

5.5. Time and Duration of Component D

process. *Shibuya and Sasajima* [1986] found that the polarity boundary is parallel to the bedding plane and conclude that the magnetization was acquired sequentially. *Fujii et al.* [1993] also conducted magnetostratigraphic work with the magnetization of component D whose polarities were consistent between the two limbs of a mesoscopic fold.

Bedded cherts of the Inuyama area show an alternation of chert layer (1.5~15 cm on average) and thinner siliceous claystone layer (1~20 mm). *Hori et al.* [1993] estimated the depositional rate for both chert and shale layers of each bed. They counted magnetic microspherules per volume in each bed, which are assumed to be extraterrestrial particles constantly supplied to the sediments. They concluded that the sedimentation rates for the chert layer and the siliceous claystone layer are 15 mm/kyr and ~0.15 mm/kyr, respectively. The constantly supplied particles composing siliceous claystone were diluted in chert layer by relatively abundant siliceous materials such as radiolaria. The sedimentation rate implies that each paleomagnetic specimen of 22-mm height represents the time period of about 1500 years. This is enough to partially average out the secular variation of the geomagnetic field. The slowly progressing diagenesis may have played an additional role in integrating the geomagnetic field directions for longer periods.

5.6. Paleolatitude at Time of Deposition

After the plate reconstruction of the Pacific by *Engebretson et al.* [1985], *Nakanishi et al.* [1992a, b] made significant progress in the identification of magnetic lineations from the western Pacific. *Nakanishi et al.* [1992b] suggested that the growth of the Pacific plate began at about 190 Ma in Early Jurassic. From the D component of magnetization, we can calculate the paleolatitude of the Izanagi plate on which the chert was deposited. Paleolatitudes calculated for all sites are listed in Table 7. Site KA6 yielded only three reliable sample directions, and thus paleolatitude was not calculated for this site. As in the previous sections, six sites passed the fold tests according to the mesoscopic fold, and the shallowing of the inclinations is corrected. The results are plotted in Figure 19 versus age estimated from radiolarian fossils. A line was fitted by linear regression method considering the weight of the error in the paleolatitude at each site, which has a slope of -0.50 ± 0.44 deg/Myr. This corresponds to the minimum migration speed of 5.6 ± 4.9 cm/yr. The best estimate of the paleolatitude for each age is obtained by reading from Figure

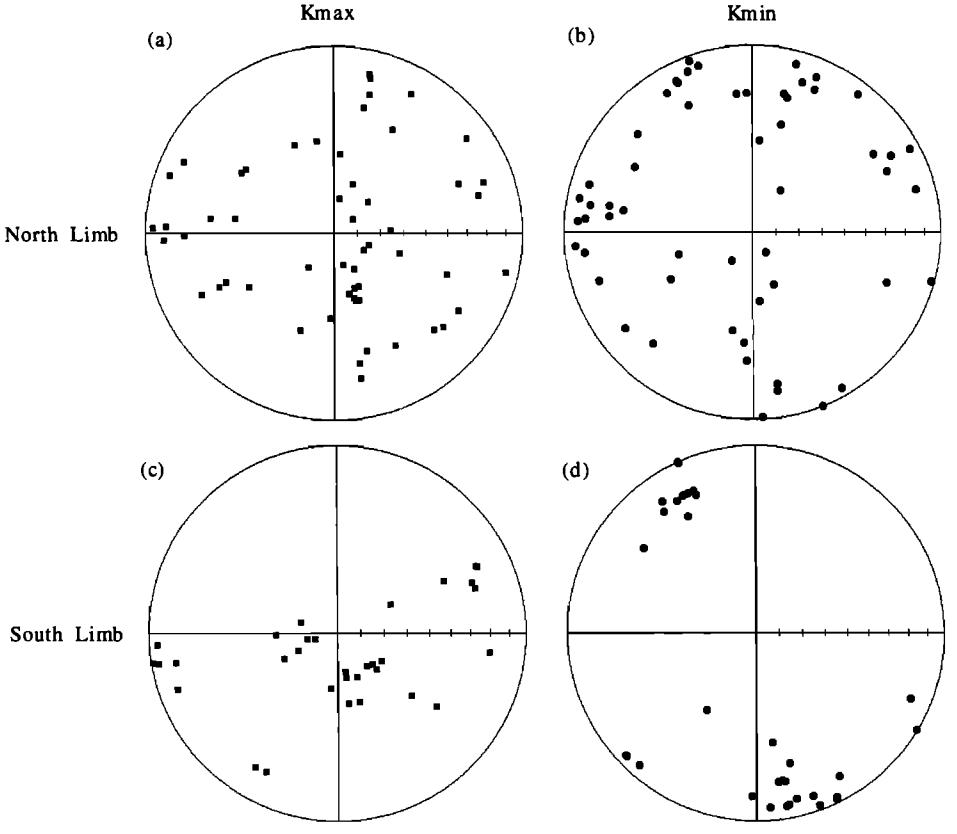


Figure 15. Insitu directions of (a) K_{max} and (b) K_{min} for the north limb of the Sakahogi Synform and of (c) K_{max} and (d) K_{min} for the south limb.

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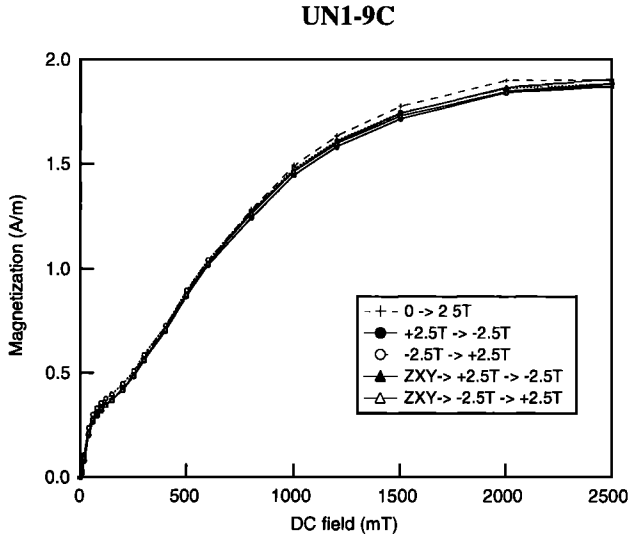


Figure 16. IRM acquisition curves in the *Z* axis for UN1-9C. Crosses represent the acquisition curve from the demagnetized state to +2.5 T. Solid (open) circles represent stepwise back field application from +2.5 T to -2.5 T divided by 2 after subtraction of the starting remanence of each sequence. Solid (open) triangles represent the results of the same procedure repeated after the application of the same procedure in the *X* and *Y* axes. There are only little differences between each sequence except between the initial acquisition of IRM from the demagnetized state and others.

19. For example, the estimated paleolatitudes for 237 Ma, 223.5 Ma, and 210 Ma are 12.3° ± 15.6°N or S, 19.0° ± 10.5°N or S, and 25.7° ± 16.7°N or S. In the calculation we assumed that chert beds were deposited at the same latitude.

The paleolatitude obtained by *Shibuya and Sasajima* [1986] is also plotted by an open triangle in Figure 19. They used all Middle Triassic sites for the calculation of paleolatitude. The paleolatitude of 0.7° ± 3.4°N or S obtained by *Shibuya and Sasajima* [1986] is within the 95% confidence limit (2σ) but slightly lower than ours. One possible cause of the lower paleolatitude for *Shibuya and Sasajima* [1986] is that all 52 specimens which were used for the estimation of their paleolatitude come from only a single site of about 40-m thickness (site 41). They conducted bedding correction and obtained magnetizations of both polarities. However, the inclinations of individual specimens are higher than the mean inclinations calculated (1.4° ± 6.8°). Although the reason is not clear, their normal and reversed polarity directions are not completely antipodal and cancel out by averaging both polarities [see *Shibuya and Sasajima,* 1986, Figure 5]. On the other hand, our paleolatitude and magnetization direction are obtained from each site of single chert bed after unfolding along the mesoscopic fold and are used for the statistical analysis.

Recently, *Otofuji* [1996] proposed that east Asia, including the Japanese Islands, experienced a northward translation of about 1700 km between 20 Ma and 10 Ma. Their proposal is based on the Late Cretaceous paleomagnetic results from Sikhote Alin [*Otofuji et al.,* 1995] combined with other paleomagnetic data from east Asia. They suggested no significant latitudinal movement between 60 Ma and 20 Ma for east Asia. If the northward translation between 20 Ma and 10 Ma is applied to the studied area (35.4°N), a paleolatitude of about 20°N at 60 Ma can be obtained. It is believed that the Jurassic accretionary complex began to be formed at around 160 Ma [*Kimura,* 1999]. Assuming that the paleolatitude of the studied area is 20°N at 160 Ma and has a southern latitude at 210 Ma (25.7° ± 16.7°S), we get a southward migration from 237 Ma to 210 Ma and a northward migration from 210 Ma to 160 Ma at a speed of 9.1 ± 3.4 cm/yr. This is a considerably high speed of latitudinal migration persisting for about 50 Myr. If we assume northern latitude throughout, the northward migration was almost terminated at around 210 Ma.

5.7. Timing and Origin of Component C

Component C shows positive inclination (Table 3) and thus is thought to have been acquired during a normal polarity

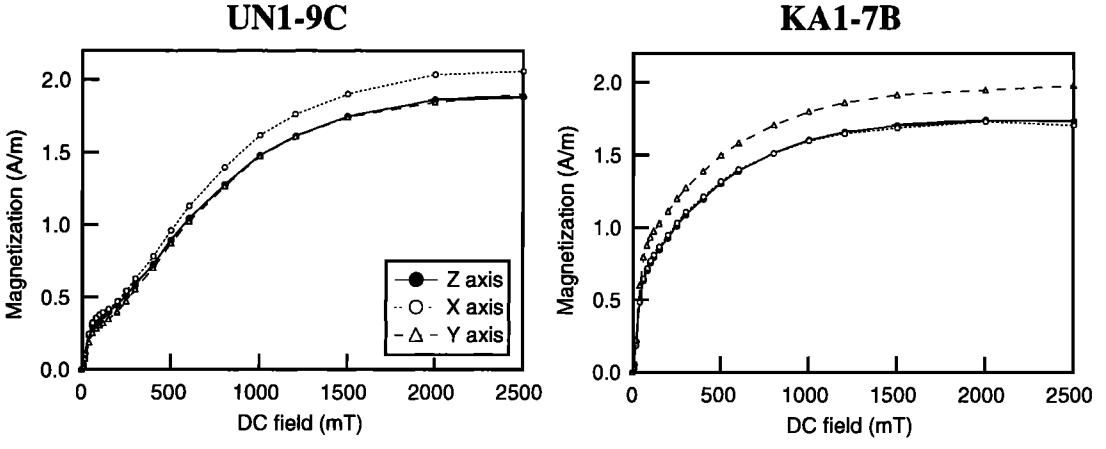


Figure 17. IRM acquisition curves (magnetization versus dc field applied) for five selected specimens from the red chert. A dc magnetic field of -2.5 T was applied to the *Z* axis and was subsequently exposed to stepwise IRM acquisition up to +2.5 T according to *Tamai* [1998]. The resulting magnetization at each step is subtracted from the final IRM at +2.5 T and divided by 2. The same procedure was repeated in the *X* and *Y* axis.

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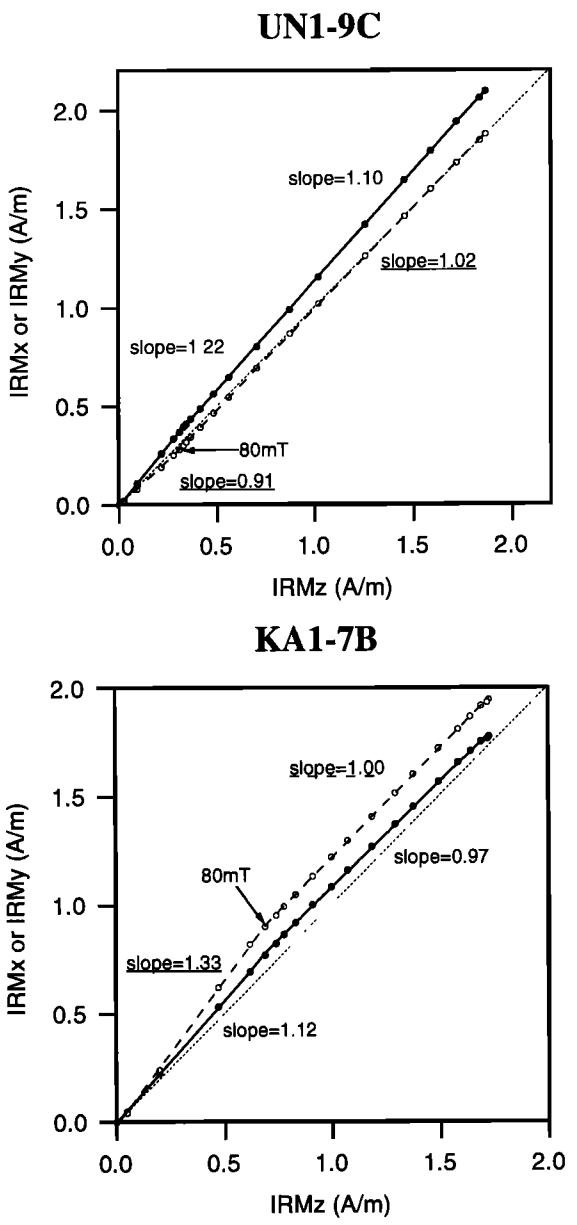


Figure 18. IRM_x (solid circles) and IRM_y (open circles) versus IRM_z plots for the steps of dc field. Data points from a perfectly isotropic rock specimen should lie on the dotted lines with a slope of unity. Samples with different slopes for lower and higher coercivities mean that magnetite (lower) and hematite (higher) have different magnetic anisotropy.

chron. The fact that component C does not pass the fold test with the mesoscopic fold indicates that component C is a secondary component which was acquired after the bending of the mesoscopic fold. The synfolding magnetization suggested by the fold tests shows that the magnetization was acquired during the formation of Sakahogi Synform. The observation of a near-perpendicular relationship between component C magnetizations and the K_{min} axis of AMS for sites UN1 and MO1 may possibly be the effect of magnetic fabric on component C. However, it is also possible that this relationship is coincidental.

Kimura [1999] recalculated the unfolding of the directions of component C reported by *Shibuya and Sasajima* [1986] by

Table 6. Magnetic Anisotropy Parameters

Sample	Susceptibility	Anisotropy of Magnetic Susceptibility							Anisotropy of Isothermal Remanent Magnetization						Inclination	Corrected Inclination	
					Maximum		Minimum		Low Coercivity (0-80 mT)			High Coercivity (80-2500 mT)			After	Inclination	
	($\times 10^{-6}$ SI)	P_J	T	Shape	Dec	Inc.	Dec.	Inc.	I_x/I_z	I_y/I_z	shape	I_x/I_z	I_y/I_z	shape	I_{hmax}/I_z	Unfolding	$a = \infty$
UN1-9C	241	1 09	0.183	oblate	86	9	177	2	1.22	0.91	triaxial	1.10	1.02	prolate (X)	1.10	12.0	13.2
UN2-7B	259	1.05	0.279	oblate	85	10	277	79	0.84	0 89	prolate (Z)	1.03	1.07	triaxial	1.07	46.6	48.5
KA1-7B	240	1.04	0.860	oblate	358	58	94	4	1.12	1.33	prolate (Y)	0.97	1.00	~isotropic	1.00	22.6	22.6
KA2-8B	507	1.01	-0.614	prolate	223	15	318	19	0.99	1.00	~isotropic	1.06	1.06	oblate (Z)	1.06	39.4	41.0
KA3-8B	131	1.01	-0.861	prolate	132	1	224	52	1.12	1.07	triaxial	1.07	1.04	triaxial	1.07	16.5	17.6
KA5-1A	290	1.02	0.143	oblate	6	3	274	39	0.86	1.04	oblate (X)	1.00	1.08	prolate (Y)	1.08	29.4	31.3

Principle axes are presented in the specimin coordinates whose z axis is perpendicular to the bedding plane. I_x, I_y, and I_z are isothermal remanent magnetization in X, Y, and Z axes, respectively.

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Table 7. Paleolatitude Calculated for Each Site

Site	*N*	Polarity	Inclination	Paleolatitude (°N or °S)	Inclination Corrected	Paleolatitude Corrected (°N or °S)
UN1	9	-	12.0	6.1 ± 17.7	13.2	6.7 ± 17.8
UN2	7	R (N)	46.6	27.9 ± 16.7	48.5	29.5 ± 17.4
KA1	10	N (R)	22.6	11.8 ± 14.6	22.6	11.8 ± 14.6
KA2	9	N (R)	39.4	22.3 ± 13.8	41.0	23.5 ± 14.2
KA3	7	N (R)	16.5	8.4 ± 23.1	17.6	9.0 ± 23.3
KA5	8	N (R)	29.4	15.8 ± 10.0	31.3	16.9 ± 10.2

using an axis with a trend of N82°W plunging 78° with a rotation angle of 90°. The angle between the direction of northern and southern limbs after unfolding is 10° (Figure 20b), and the full unfolding seems to be appropriate. We tried 50% unfolding (45° rotation) for the component C direction of the northern limb for site 16 of *Shibuya and Sasajima* [1986]. The corrected direction (open rectangle; Figure 20b) is still not far away from the direction of their site 41 (solid rectangle) and within the confidence limit of the direction of the southern limb of this study (solid circle). The difference between the unfolding of the directions of *Shibuya and Sasajima* [1986] and this study can be understood as a local scattering from site to site, because *Shibuya and Sasajima* [1986] performed the unfolding of the Sakahogi Synform by using the directions of only two sites (sites 16 and 41).

Takami and Itaya [1996] obtained K-Ar ages of the fine-grained metamorphic phengite separated from the chaotic sedimentary unit of the Jurassic accretionary complex. They used the samples from the western part of the Jurassic accretionary complex in Japan and reported three different ages of 129, 152, and 169 Ma for the three different units of the area, which is 23 m.y. after the accretion of each unit. The metamorphic temperature (prehnite-pumpellyite to pumpellyite-actinolite facies) is estimated close to 350°C but no higher than that, which corresponds to the closing temperature of the K-Ar system for phengite. Although the Inuyama area is distant from the above mentioned area and is within the coherent unit, not the chaotic unit, the above information can be used for the Inuyama area. The coherent unit is considered to have experienced lower temperature than the chaotic unit. Assuming that the highest metamorphic grade is prehnite-pumpellyite facies as suggested by *Matsuda and Isozaki* [1991], the maximum temperature attained may be 200°C or less. Thus the high unblocking temperature (~500°-530°C) of component C can only be explained by chemical remanent magnetization (CRM), which was also suggested by *Shibuya and Sasajima* [1986]. Although the exact mechanism and the timing of the production of magnetite is not clear, it was probably formed as a chemical process associated with subduction. The rather high coercivity of the magnetite component suggests that the magnetite grains are in the pseudo-single-domain to single-domain range.

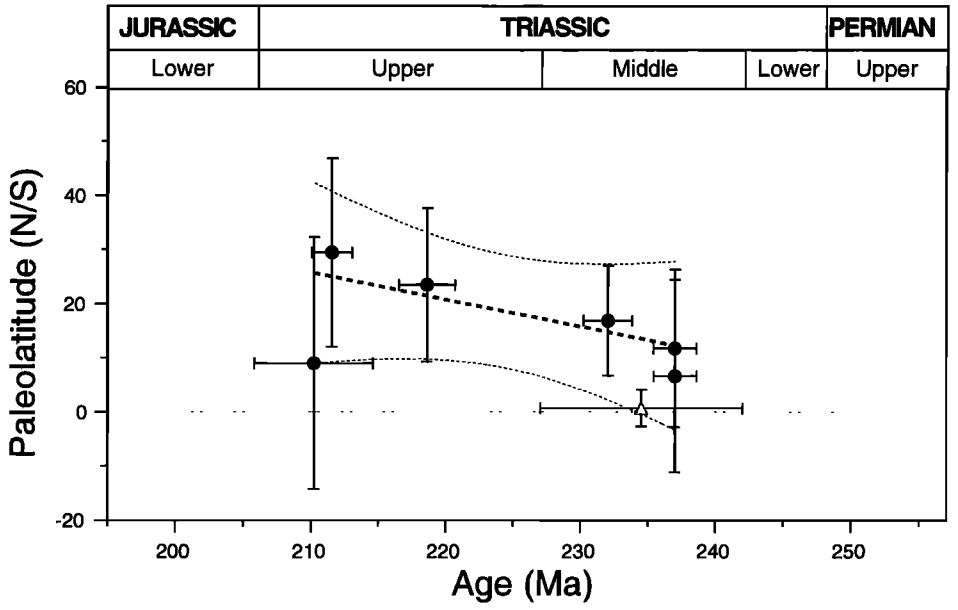


Figure 19. Paleolatitude versus age for each site which passed the fold test along the mesoscopic fold. The linear regression line (thick dashed line) fitted to the data has a slope of -0.50 ± 0.44 (2σ) deg/Myr, which corresponds to 5.6 ± 4.9 cm/yr. The 95% confidence limits of the fitted line are shown as thin dashed curves. Open triangle is a paleolatitude obtained by *Shibuya and Sasajima* [1986].

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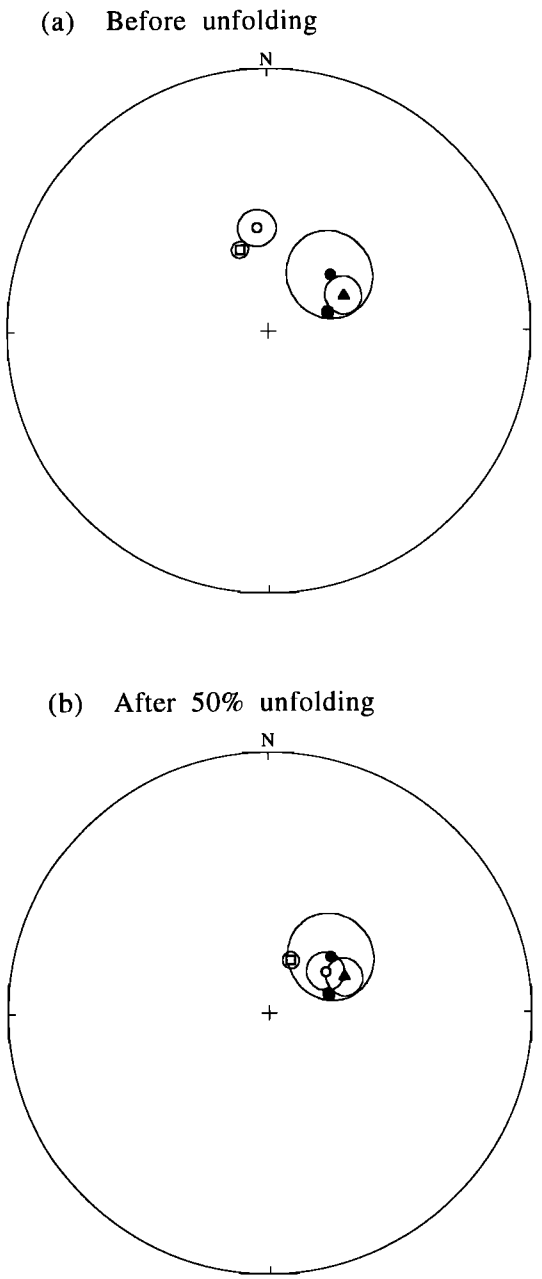


Figure 20. Equal-area projections of paleomagnetic directions on the Southern Hemisphere for component C from southern limb (solid circles) and northern limb (open circles) (a) before unfolding and (b) after 50% unfolding. Unfolding was conducted in the direction of comoponent C from northern limb with a trend of N82°W plunging 78° and the rotation angle of 110°. Paleomagnetic directions from site 16 (northern limb, open rectangles) and site 41 (southern limb, solid rectangles) of *Shibuya and Sasajima* [1986] are shown for comparison. Paleomagnetic direction from SW Japan for 85-112 Ma by *Otofuji and Matsuda* [1987] is also shown as a solid triangle.

The Cretaceous (112-85 Ma) paleomagnetic direction of southwest Japan [*Otofuji and Matsuda,* 1987] is also plotted in Figure 20 (solid triangle). The direction agrees well with the directions of component C after unfolding (Figure 20b). Considering that the polarity of the magnetization for component C is normal and that its direction agrees well with the direction of SW Japan for 112-85 Ma [*Otofuji and Matsuda,* 1987], the time of magnetization acquisition is considered to be during the Cretaceous Normal Superchron (120.5-83.5 Ma [*Gradstein et al.,* 1995]). These lines of evidence lead to the conclusion that during the Cretaceous the Inuyama area moved together with SW Japan after the formation of the Sakahogi Synform. The movement of SW Japan includes a northward translation of 1700 km, accompanied by a clockwise rotation of 10° of east Asia, and a subsequent 45° clockwise rotation of SW Japan is associated with the opening of the Japan Sea at 15 Ma [*Otofuji,* 1996]. *Kimura* [1999] suggested a post-Miocene 15° counterclockwise rotation associated with the bending of the eastern part of SW Japan during collision of the Izu-Ogasawara (Bonin) arc.

5.8. Timing and Origin of Component B

A remarkable feature of component B is that all sites show negative inclination. Although the site mean directions are somewhat scattered especially for the sites from the southern limb of the synform, the directions point ESE in general after inverting the polarity (Table 3). As there is no difference between the site mean directions from the northern and southern limbs, component B is thought to have been acquired after the bending of the Sakahogi Synform. The large deviation of the directions in the southern limb may be related to the higher deformation inferred from the higher degree of AMS for the southern limb (Figure 13a). The mean direction of component B from all the sites is declination = -62.1°, inclination = -59.0°, α_{95} = 21.1°. The overlapping coercivity range of component B with that of component C (Figure 5a), in combination with the sharp separation by PThD at a low unblocking temperature of about 350°C (Figure 5b), suggests that component B was acquired as a thermoviscous remanent magnetization [*Dunlop and Özdemir,* 1990]. This means that the area was heated up to a few hundred degrees for a geologically long period to erase the low unblocking fraction of the component C magnetization carried by magnetite. However, the maximum temperature was low enough to preserve the higher unblocking fraction of component C.

The mean direction of component B is plotted in Figure 21 together with the directions from SW Japan by *Otofuji and Matsuda* [1987] for three periods: 80-92 Ma (solid rectangle), 58-63 Ma (shaded rectangle), and 20-30 Ma (open rectangle). Although the mean B direction differs somewhat from the directions of SW Japan, its α_{95} circle overlaps that of the 58-63 Ma mean direction. According to the Watson test [*Watson,* 1956], the statistic F was calculated as 4.64, which means that the two data sets have distinct directions with a rejection limit of 5% (3.22) but are not distinct with a rejection limit of 1% (5.15). In southwest Japan, large granitic batholiths of Late Cretaceous age are well distributed. The age of the granites coincides with the subduction of a ridge between the Izanagi and Pacific plates in the Late Cretaceous [*Kinoshita,* 1995]. The K/Ar age of the granites becomes younger from 90 Ma to 60 Ma eastward, reflecting the eastward migration of the ridge along the trench [*Kinoshita,* 1995]. The K/Ar age of the granitic bodies around the studied area ranges from 65 to 75 Ma [*Kinoshita,* 1995; *Sato et al.,* 1992]. Component B is estimated to be acquired as a secondary thermoviscous remanent magnetization which was initiated by heating associated with the ridge subduction and terminated by cooling related to

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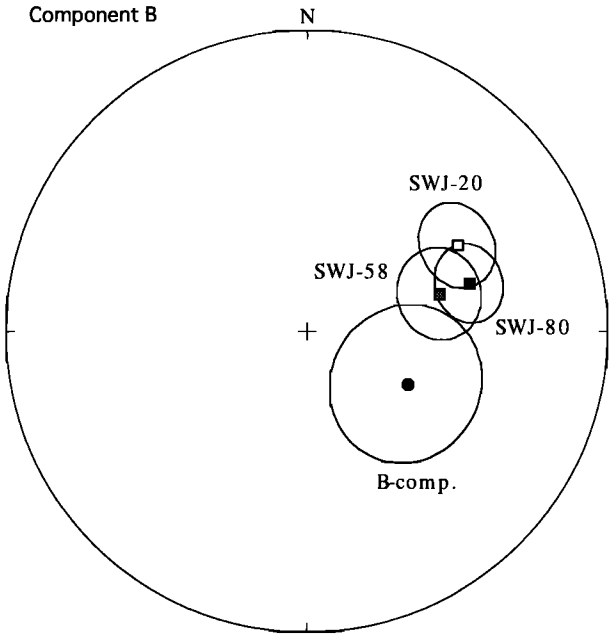


Figure 21. Equal-area projection of mean paleomagnetic direction in the Southern Hemisphere for component B from all the sites (solid circle). Paleomagnetic directions from SW Japan for 80-92 Ma (solid rectangle), 58-63 Ma (shaded rectangle), and 20-30 Ma (open rectangle) by *Otofuji and Matsuda* [1987] are also shown.

the subsequent uplift and erosion of the accretionary complex. There are two possible periods of time for the blocking of component B with reversed polarity. One is Chron C31r which lasted about 2.3 Myr from 71 Ma and intervened between relatively long normal chrons [*Gradstein et al.*, 1995]. Another is the period between Chrons C27r and C24r (53.3-62.5 Ma), which has dominantly reversed polarity [*Berggren et al.*, 1995]. The latter is favored because the paleomagnetic direction from SW Japan is closer for the period 58-63 Ma mentioned above (Figure 21). According to the latter hypothesis, the formation of the granitic batholiths was around 65-75 Ma after passage of the ridge between the Izanagi and Pacific plates, and the subsequent cooling was terminated between 62.5 Ma and 53.3 Ma, resulting in the thermoviscous remanent magnetization of reversed polarity.

Fission track dating on a Jurassic accretionary complex in Japan is currently taking place [*Watanabe and Hasebe*, 1999]. The preliminary results show that fission track ages of zircon are subdivided into three groups: i.e., >130 Ma (8 sites), 80-120 Ma (15 sites), and 50-60 Ma (4 sites). They interpreted the age older than 80 Ma to be due to uplift of the area and that of 50-60 Ma to be due to reheating by granitic batholith intrusions. Generally, ages older than 80 Ma become younger from north to south depending on the distance from the Ryoke metamorphic belt. The dominance of the zircon fission track ages (closure temperature of 250°C) around 80-120 Ma supports the idea that component C was acquired during the Cretaceous Normal Superchron.

Younger ages (50-60 Ma) at four sites indicate that some areas closer to the heat source, such as granitic bodies, have been heated to the closure temperature of fission tracks of zircon (250°C), whereas other places are less than 250°C. Apatite fission track ages show about 60 Ma on all 10 sites detected, which means that the Jurassic accretionary complex is uplifted, eroded, and cooled down to the closure temperature of apatite (~80°C) around 60 Ma. This is consistent with our interpretation that the component B magnetization was acquired between 62.5 Ma and 53.3 Ma.

Based on the theoretical blocking curves for magnetite by *Pullaiah et al.* [1975], approximate temperature during heating by the granitic batholiths was estimated. Assuming the constant temperature and the duration of 10 Myr, we obtain 200°C for the unblocking temperature of component B (350°C). As the closure temperature of zircon (250°C) is close to but slightly above the estimated average temperature (200°C), the above mentioned zircon fission track ages of two groups (80-120 Ma and 50-60 Ma) seem consistent with our results.

6. Conclusions

A paleomagnetic study of Triassic and Jurassic red bedded cherts distributed in the Inuyama area has provided an excellent case study for the tectonic history of the accretionary complex, supported by detailed age determinations from radiolarians. Four magnetic components isolated can be interpreted in terms of the tectonic history of the red bedded chert, from the initial deposition, through accretion and bending of the Sakahogi Synform, to erosion and exposure at the surface. Figure 22 illustrates the tectonic history of the Inuyama area including our paleomagnetic results and other geological and tectonic evidence.

Component D has both normal and reversed polarities carried by hematite with unblocking temperatures of 560°-680°C. Six out of nine sites from Middle to Late Triassic yielded stable magnetization directions which pass the fold test with the mesoscopic fold. The anisotropy of isothermal remanent magnetization was measured, and the maximum inclination shallowing of about 2° was estimated and corrected for the calculation of the paleolatitudes. Paleolatitude was obtained as 12.3° ± 15.6°N or S, 19.0° ± 10.5°N or S, and 25.7° ± 16.7°N or S for 237 Ma, 223.5 Ma, and 210 Ma, respectively. Assuming Southern Hemisphere latitudes, a southward migration of 5.6 ± 4.9 cm/yr was estimated between 237 Ma and 210 Ma, and northward migration of 9.1 ± 3.4 cm/yr was estimated from 210 Ma to 160 Ma. In the case of northern latitudes, northward migration of 5.6 ± 4.9 cm/yr occurred between 237 Ma and 210 Ma, with no significant latitudinal migration from 210 Ma to 160 Ma.

Component C has normal polarity, which is carried by magnetite with an unblocking temperature of 350°-530°C. Component C is considered to be acquired as a synfolding CRM during the formation of the Sakahogi Synform, which is synchronous with the Ryoke low P-T metamorphism and the granitic batholith intrusions caused by the ridge subduction. The paleomagnetic direction of component C agrees well with that of SW Japan for 112-85 Ma [*Otofuji and Matsuda*, 1987], suggesting the acquisition of the magnetization during the Cretaceous Normal Superchron (120.5-83.5 Ma).

Component B has reversed polarity and the remanence-carrying mineral is magnetite with unblocking temperature of 200°-350°C. Component B is thought to be acquired as a thermoviscous remanent magnetization after the formation of the Sakahogi Synform. The reversed polarity suggests a relation with granitic batholith intrusion of Late Cretaceous (75-

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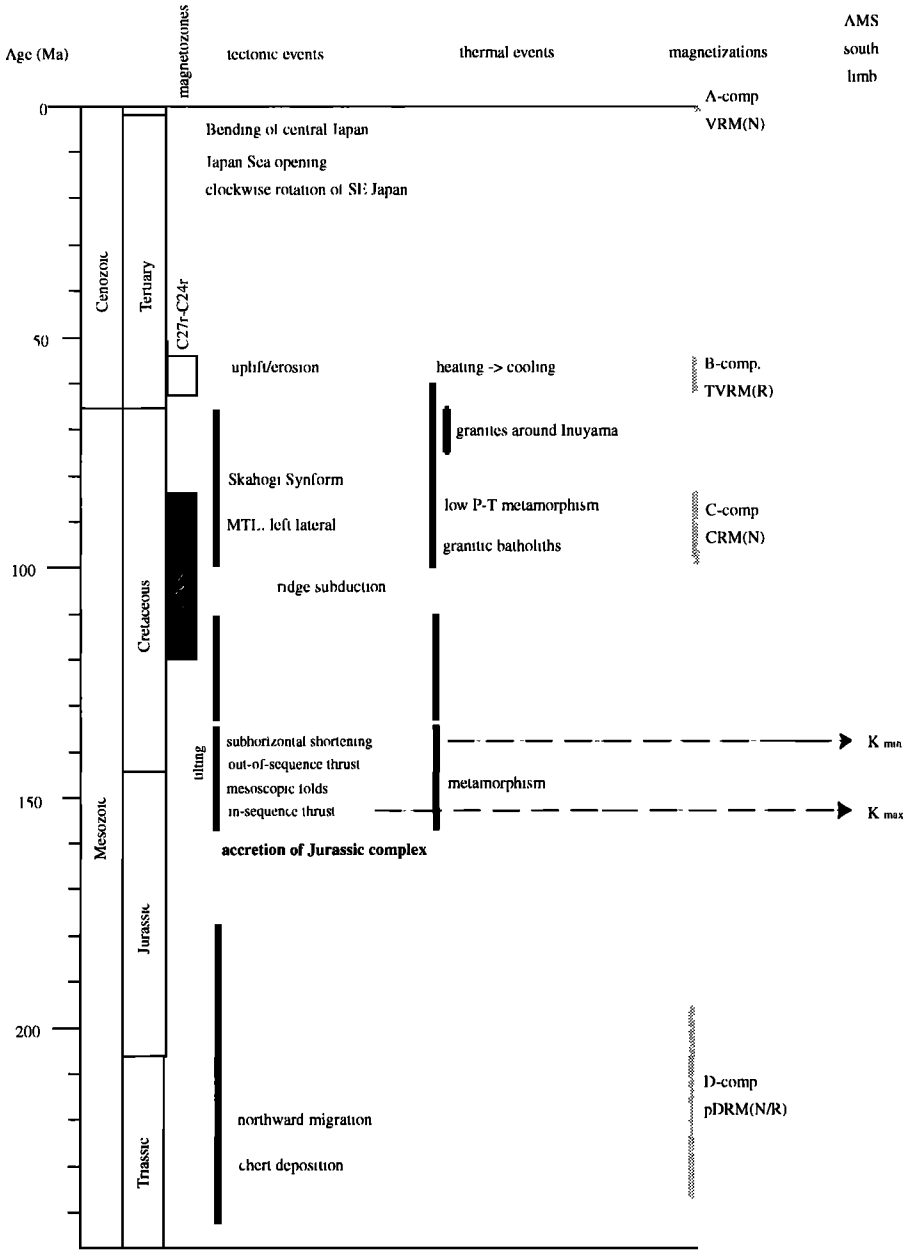


Figure 22. Schematic diagram of the geologic history of the chert sequences exposed around the Inuyama area.

65 Ma), because the magnetization was dominantly reversed between C27r and C24r (58-63 Ma). Zircon fission track ages of 50-60 Ma from the surrounding area support the idea. The time-temperature relationship suggests acquisition at an average temperature of 200°C over a 10-Myr interval.

The two sites (UN1 and MO1) from the south limb of the Sakahogi Synform have a stronger degree of anisotropy of magnetic susceptibility, which revealed that K_{max} axes are developed prior to the formation of mesoscopic fold, whereas K_{min} axes are developed during the later stage of the mesoscopic fold formation by the subhorizontal shortening perpendicular to the mesoscopic fold axes.

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